Information Visualization

Theme 2. Data Abstraction and Representation

Data Abstraction

Datasets

Data Types

- Items
- Attributes
- Links
- Positions
- Grids

Data and Dataset Types

Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

Attributes

Attribute Types

- Categorical



- Ordered
  - *Ordinal*



  - *Quantitative*



[Mun]

Data Abstraction

- Data Types
  - Items
  - Attributes
  - Links
  - Positions
  - Grids

Student	Major	Year of Study	Satisfaction Level	Exam Score	Study Hours	Passed (Yes/No)	Friends Count	Origin Country	Favorite Color
Alice	Comp.	1	High	82	15	Yes	2	USA	Blue
Bruno	Biology	3	Medium	74	10	Yes	3	Brazil	Green
Carla	History	2	Low	65	8	No	2	UK	Red
David	Math	4	Medium	78	12	Yes	3	Canada	Yellow
Emma	Physics	2	High	90	20	Yes	2	USA	Purple

Data Abstraction

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  - Items
  - Attributes
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  - Grids

Student	Major	Year of Study	Satisfaction Level	Exam Score	Study Hours	Passed (Yes/No)	Friends Count	Origin Country	Favorite Color
Alice	Co				15	Yes	2	USA	Blue
Bruno	Bio				10	Yes	3	Brazil	Green
Carla	History	2	Low	65	8	No	2	UK	Red
David	Math	4	Medium	78	12	Yes	3	Canada	Yellow
Emma	Physics	2	High	90	20	Yes	2	USA	Purple

Each row is an individual item (entity)

Data Abstraction

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Bruno	Biology	3	Medium	74	10	Yes	3	Brazil	Green
Carla	History	2	Low	65	8	No	2	UK	Red
			Medium	78	12	Yes	3	Canada	Yellow
			High	90	20	Yes	2	USA	Purple

Attributes describe properties of each student

Data Abstraction

- Data Types
  - Items
  - Attributes
  - **Links**
  - Positions
  - Grids

Student	Friends List	Major	Year of	Satisfaction Level	Exam Score	Study Hours	Passed (Yes/No)	Friends Count	Origin Country	Favorite Color
Alice	Bruno, Carla			High	82	15	Yes	2	USA	Blue
Bruno	Alice, David, Emma							3	Brazil	Green
Carla	Alice, David							2	UK	Red
David	Bruno, Carla, Emma							3	Canada	Yellow
Emma	Bruno, David			High	90	20	Yes	2	USA	Purple

Represents relationships between items (students connected to other students)

Data Abstraction

- Data Types
  - Items
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  - Links
  - Positions
  - Grids

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Explicit placement in space

Data Abstraction

Students are arranged in rows/columns; the table provides a structured view for comparison

➔ Links ➔ Positions ➔ Grids

Student	Major	Year of Study	Satisfaction Level	Exam Score	Study Hours	Passed (Yes/No)	Friends Count	Origin Country	Favorite Color
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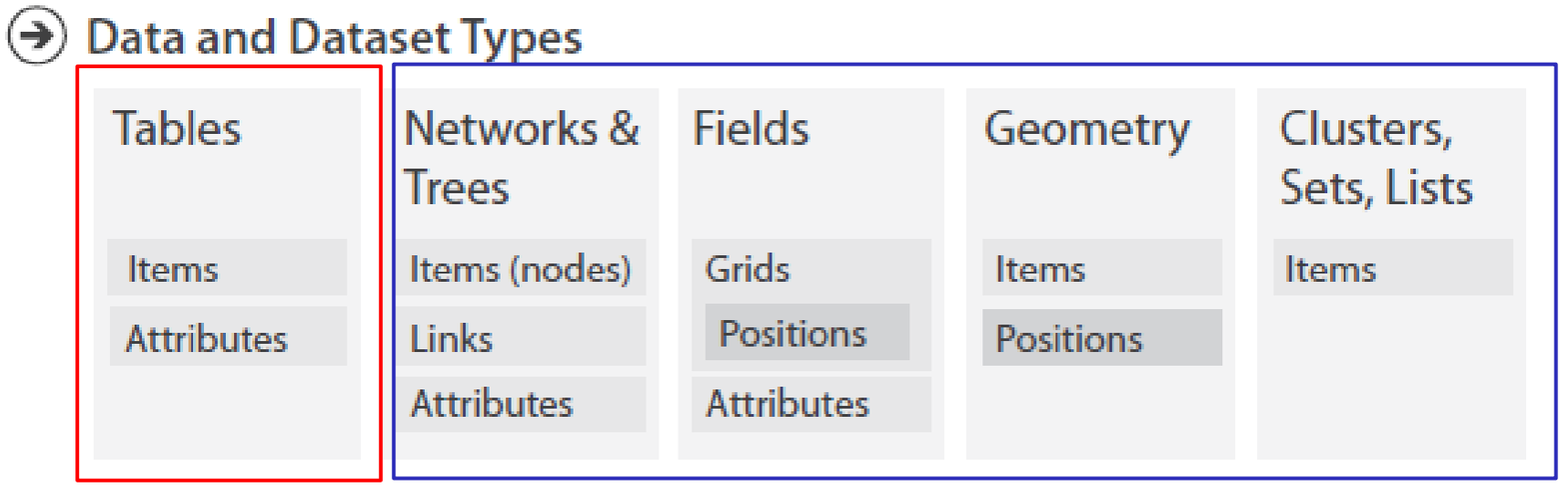
Data Abstraction

Data and Dataset Types

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Items	Items (nodes)	Grids	Items	Items
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	Attributes	Attributes		

[Mun]

Data Abstraction

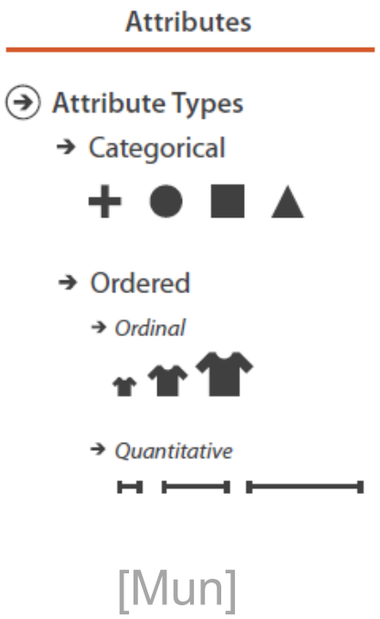


4DV805, 4DT909
Information Visualization

4DV806, 4DT912
Advanced Information Visualization

[Mun]

Data Abstraction



Data Type	Definition	Examples	Operations
Categorical	Labels without inherent order	Fruit type (apple, banana, grape), Country, Eye color	=, ≠ (compare categories)
Ordinal	Categories with a meaningful order, but no fixed scale	Survey ratings (poor, fair, good, excellent), Education level, Medal (gold, silver, bronze)	=, ≠, >, < (but not meaningful differences)
Quantitative	Numeric values with measurable differences and ratios	Temperature, Height, Income, Age	=, ≠, >, <, +, −, ×, ÷

Data Abstraction

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Emma	Physics	2	High	90	20	Yes

Categorical **Ordinal** **Quantitative**

Data Abstraction

How to transform to Ordinal?

Student	Major	Year of Study	Satisfaction Level	Exam Score	Study Hours/Week	Passed (Yes/No)
Alice	Comp.	1	High	82	15	Yes
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David	Math	4	Medium	78	12	Yes
Emma	Physics	2	High	90	20	Yes

Categorical **Ordinal** **Quantitative**

Data Abstraction

Student	Major	Year of Study	Satisfaction Level	Exam Score	Study Hours/Week	Passed (Grades)
Alice	Comp.	1	High	82	15	A
Bruno	Biology	3	Medium	74	10	C
Carla	History	2	Low	65	8	F
David	Math	4	Medium	78	12	D
Emma	Physics	2	High	90	20	B

Categorical **Ordinal** **Quantitative**

Data Abstraction

How to transform to Ordinal?

Student	Major	Year of Study	Satisfaction Level	Exam Score	Study Hours/Week	Passed (Grades)
Alice	Comp.	1	High	82	15	A
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Carla	History	2	Low	65	8	F
David	Math	4	Medium	78	12	D
Emma	Physics	2	High	90	20	B

Categorical **Ordinal** **Quantitative**

Visualizing Distributions

Example Data: Exam Scores (between 0 and 100)

72, 65, 88, 30, 45, 55, 77, 18, 68, 93,
0, 61, 74, 85, 47, 53, 79, 81, 69, 71,
25, 87, 5, 50, 58, 62, 10, 83, 29, 97

Sort Ascending

0, 5, 10, 18, 25, 29, 30, 45, 47,
50, 53, 55, 58, 61, 62, 65, 68, 69,
71, 72, 74, 77, 79, 81, 83, 85, 87,
88, 93, 97

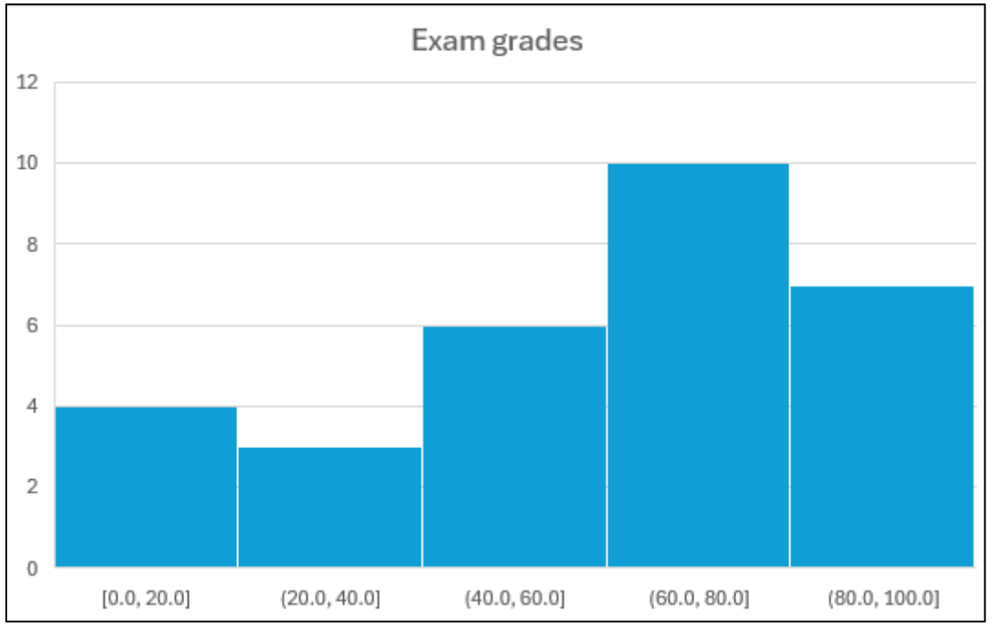
Bin Range	Frequency	Values in Bin
0–19	4	0, 5, 10, 18
20–39	3	25, 29, 30
40–59	5	45, 47, 50, 53, 55, 58
60–79	8	61, 62, 65, 68, 69, 71, 72, 74, 77, 79
80–99	10	81, 83, 85, 87, 88, 93, 97

Histograms

Example Data: Exam Scores (between 0 and 100)

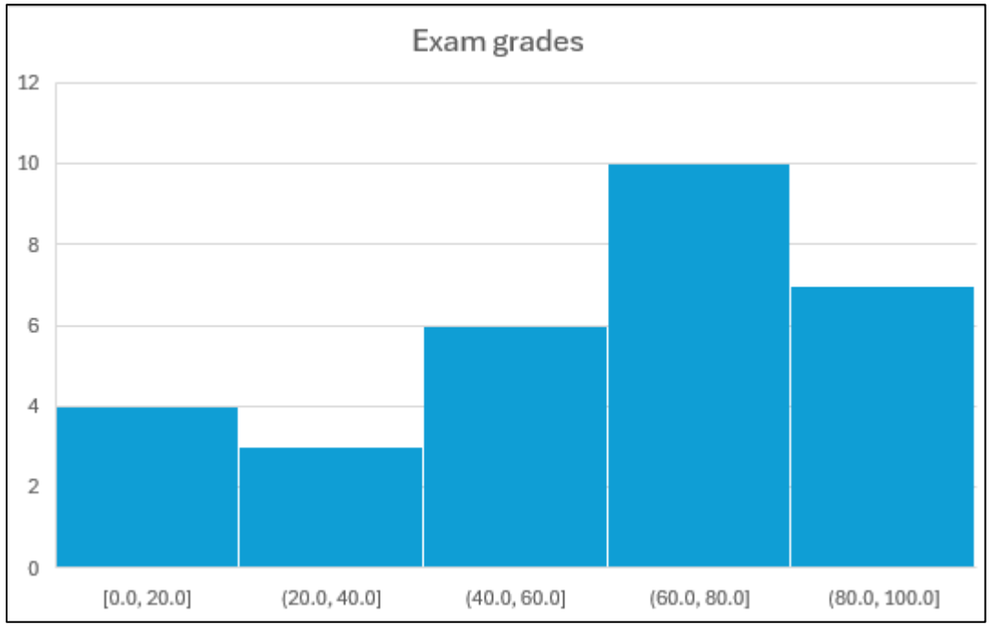
Bin Range	Frequency
0–19	4
20–39	3
40–59	5
60–79	8
80–99	10

Histogram

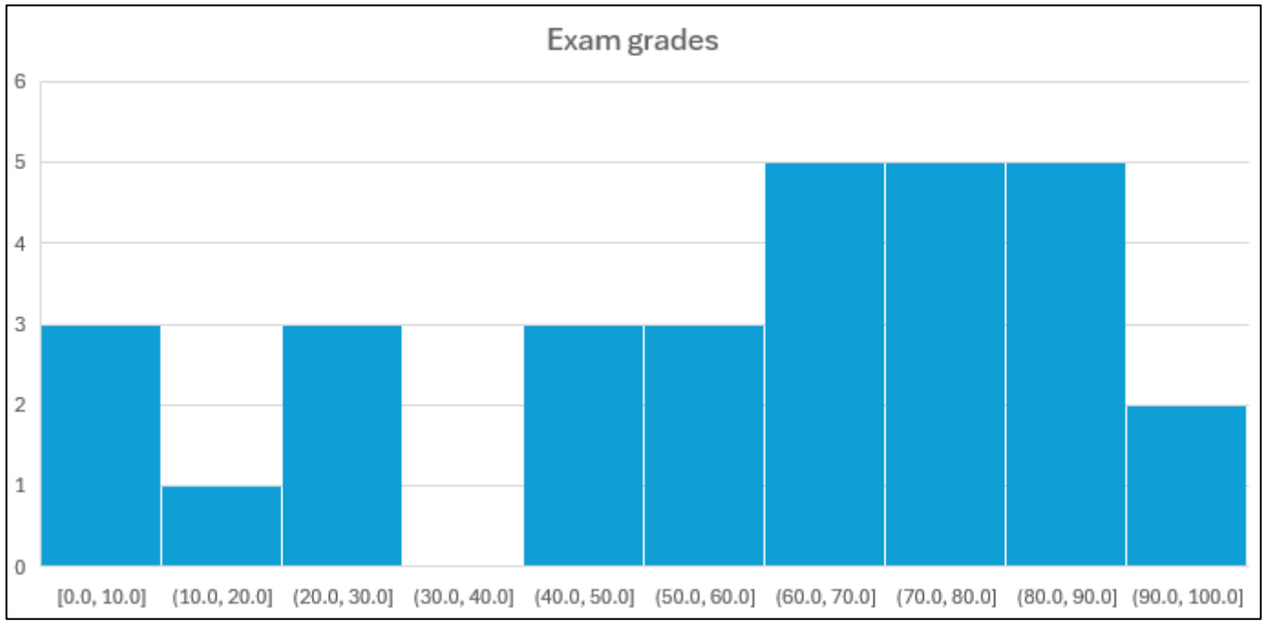


Bin Size = 20

Histograms

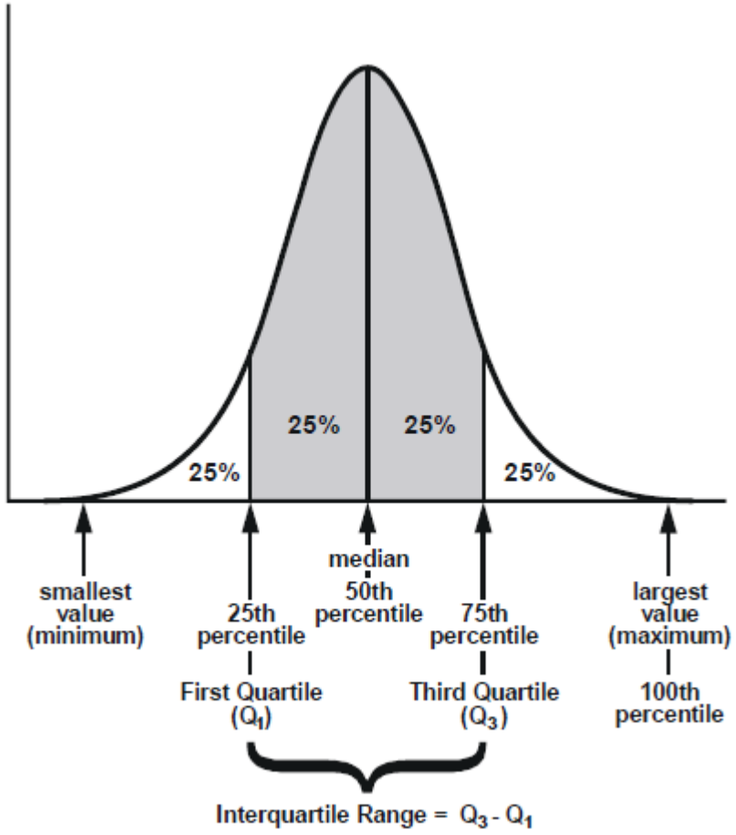


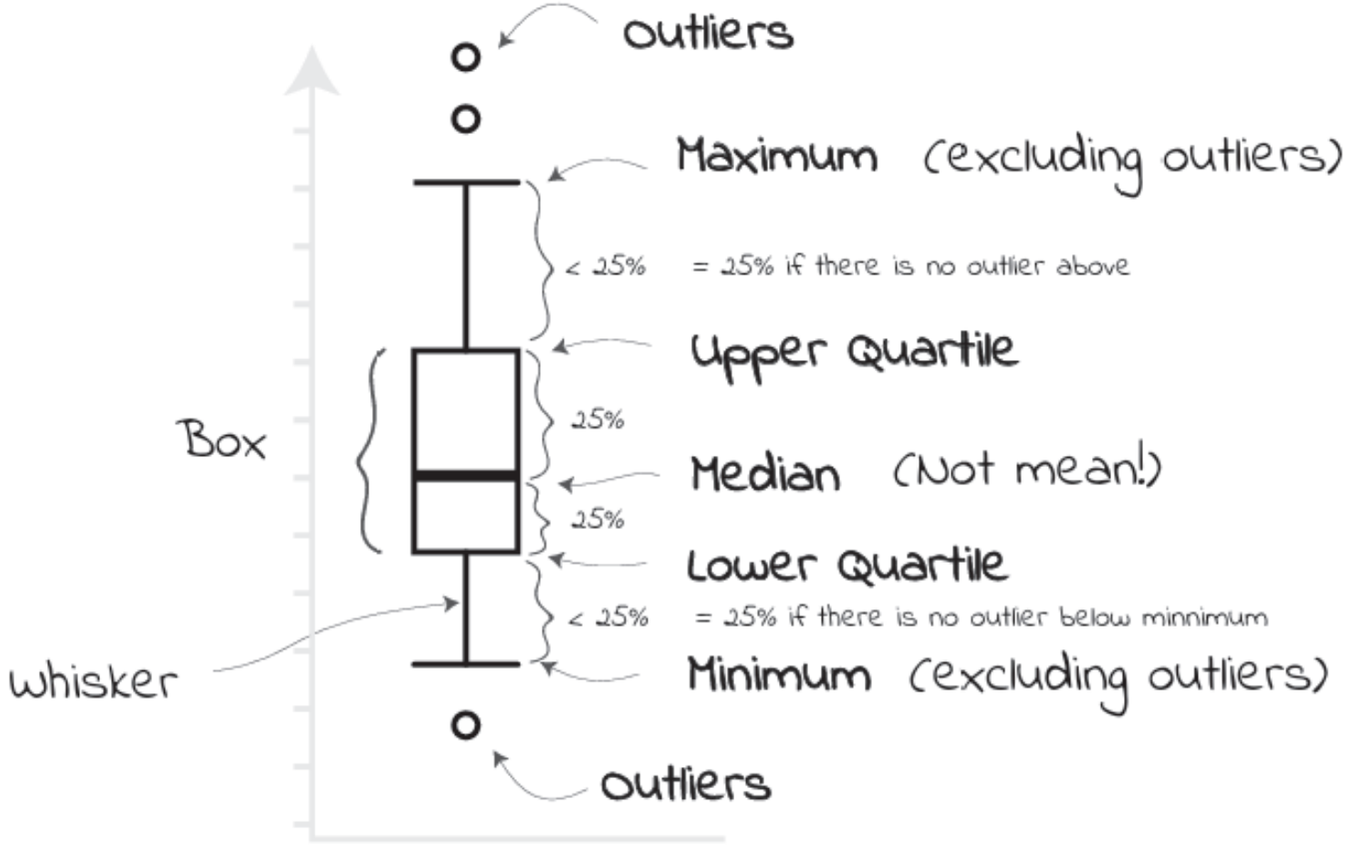
Bin Size = 20



Bin Size = 10

Visualizing Distributions

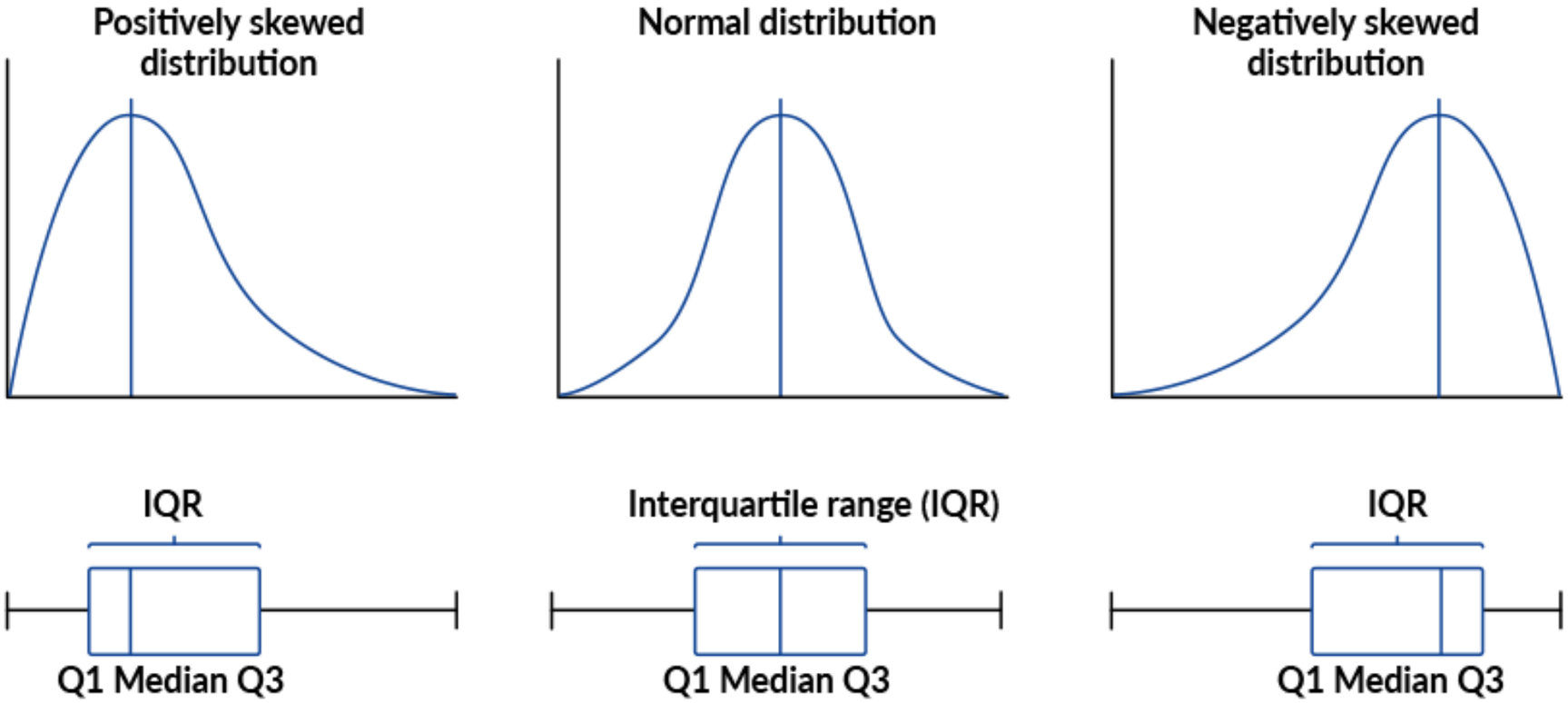




https://makemeanalyst.com/explore-your-data-range-interquartile-range-and-box-plot/

Boxplot | visualizationcheatsheets.github.io

Visualizing Distributions



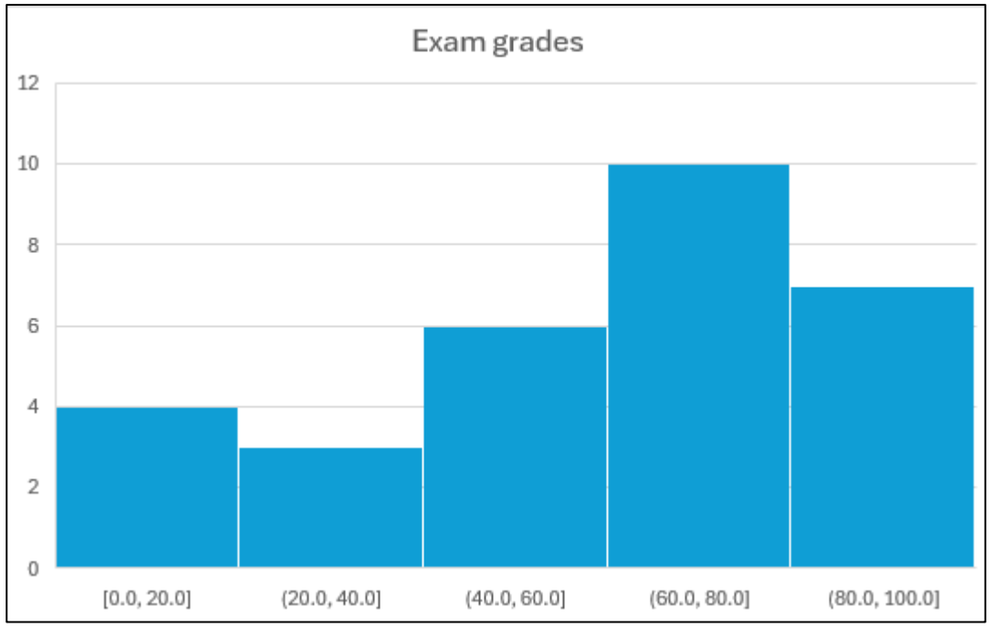
https://www.bachelorprint.com/uk/statistics/interquartile-range/

Visualizing Distributions

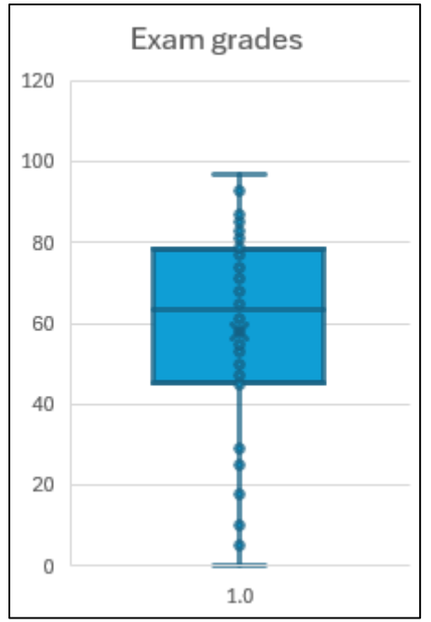
Data

0, 5, 10, 18, 25,
29, 30, 45, 47, 50,
53, 55, 58, 61, 62,
65, 68, 69, 71, 72,
74, 77, 79, 81, 83,
85, 87, 88, 93, 97

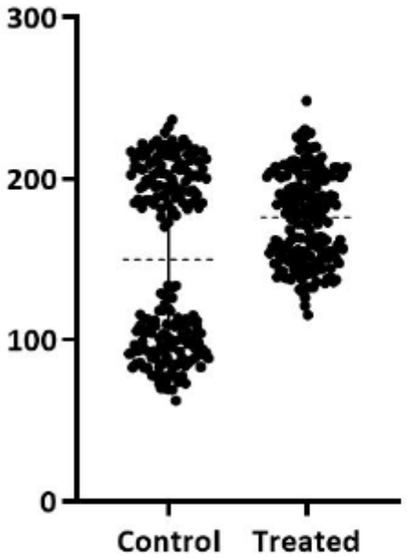
Histogram



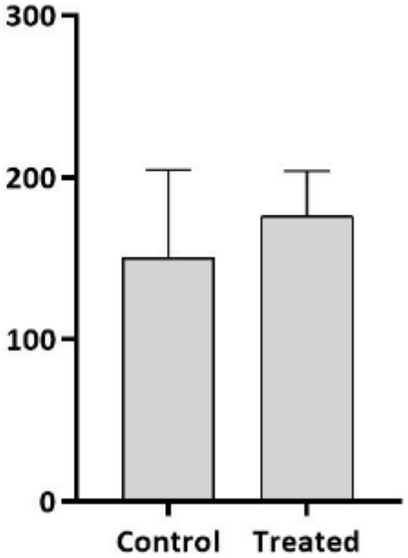
Box plots



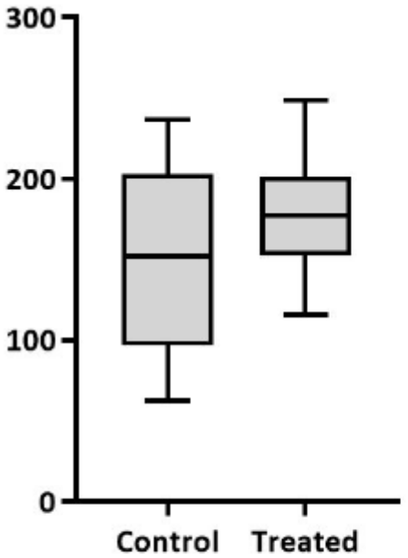
Visualizing Distributions



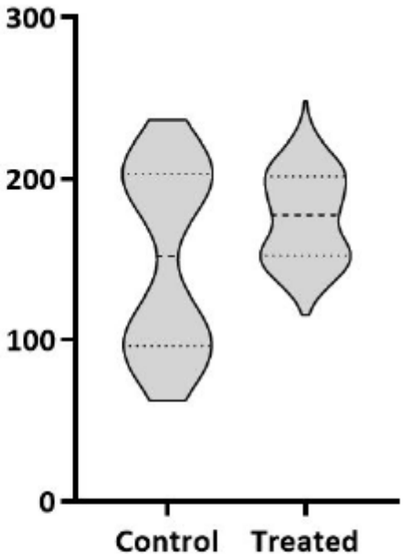
Dot plot/
Beeswarm plot



Bar charts +
error bars



Box plots

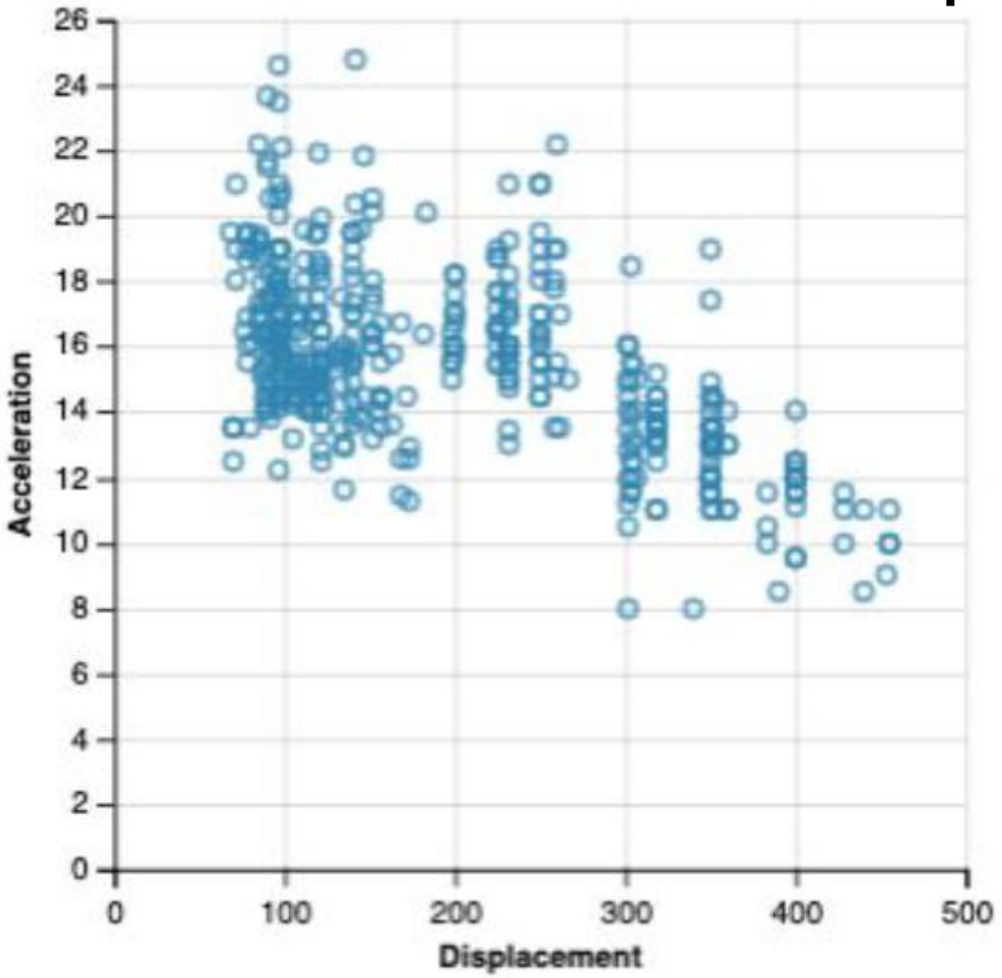


Violin plots

[Inspired by Benjamin Bach's lecture]

Visualizing Distributions

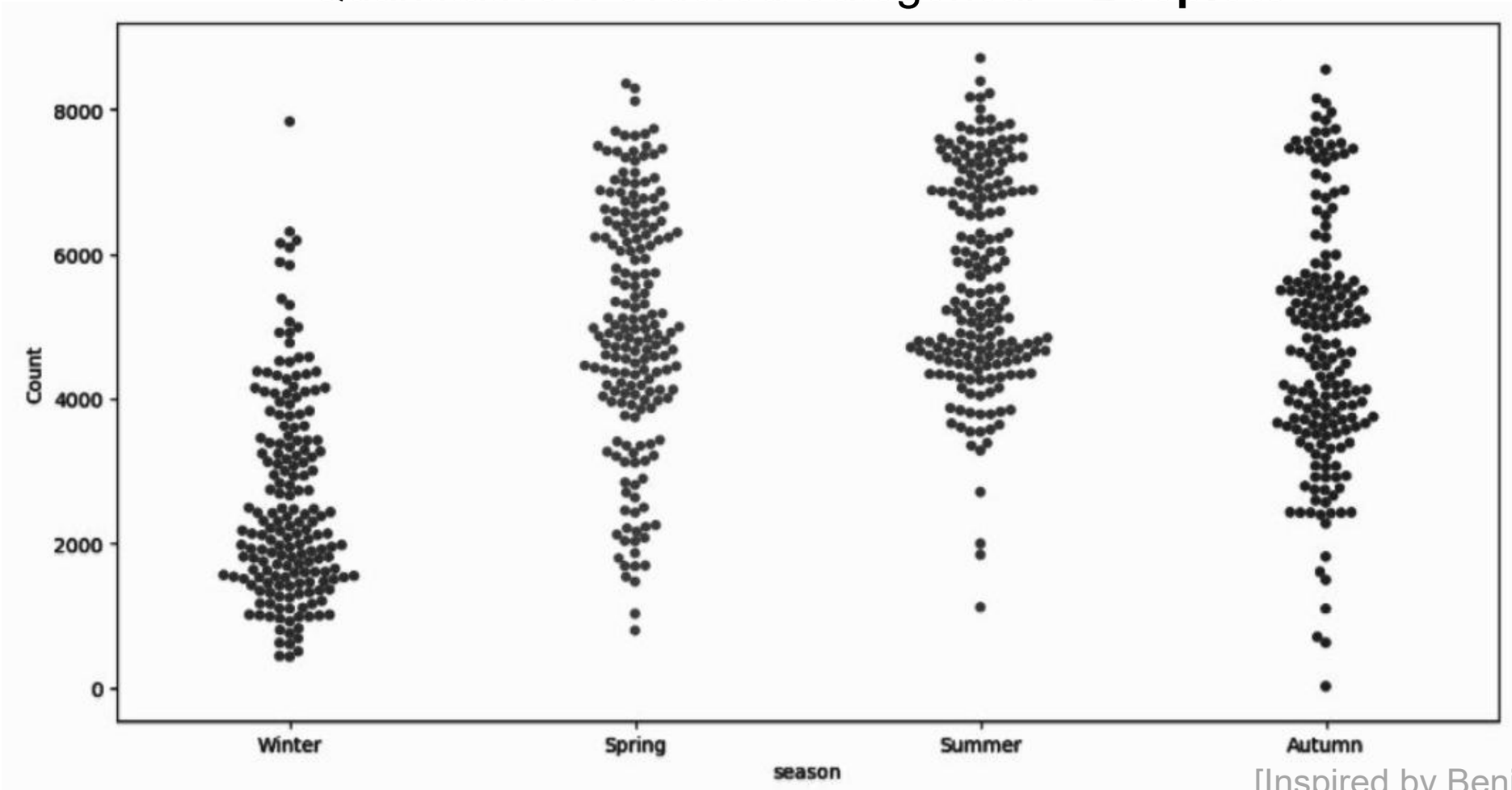
Quantitative x Quantitative = **Scatterplots**



[Inspired by Benjamin Bach's lecture]

Visualizing Distributions

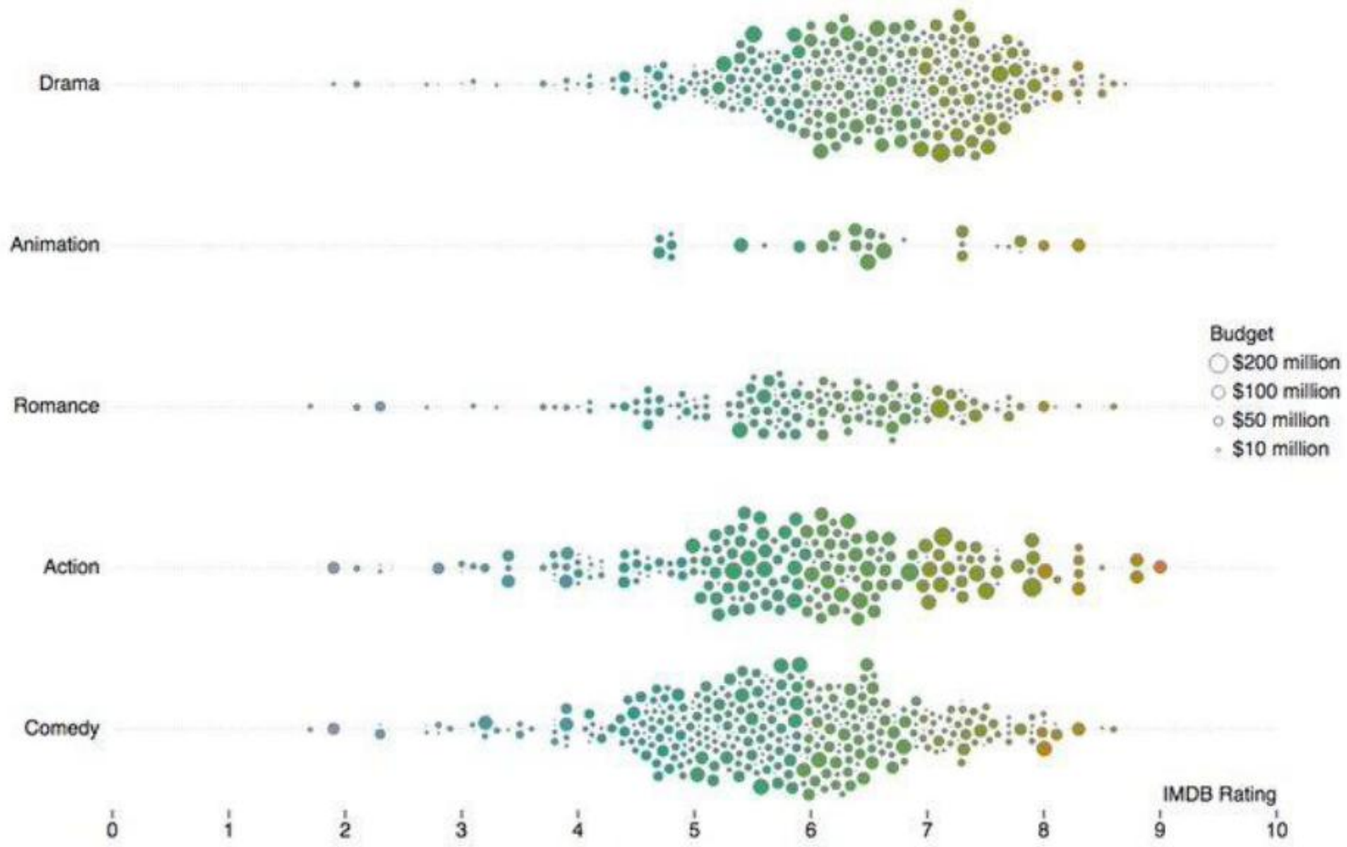
Quantitative x Ordered/Categorical = **Beeplots**



[Inspired by Benjamin Bach's lecture]

Visualizing Distributions

Quantitative x Categorical = **Swarm plots**



[Inspired by Benjamin Bach's lecture]

Data Representation

- Dimensionality (of data)
  - Depends on the number of attributes (variables)
  - 1D, 2D, 3D
  - ≥4D (nD)

- In InfoVis, we try to map all input data to 2D (or sometimes 3D) (visual mapping) in order to create a **visual encoding** of the data items that results in a final **visual representation/structure**

[Spe:34ff]

Data Representation

- One-dimensional (1D): linear data that includes textual documents, source code, list of names, all organized sequentially.

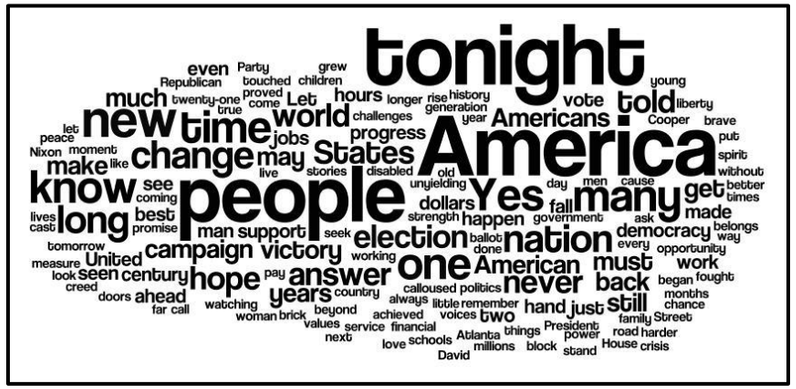
"A picture is worth a thousand words"

```
V = [3, 4, 2, 13, 23, 16]
V.sort()
print(V)
```

[The eyes have it: A task by data type taxonomy for information visualization, Shneiderman 96]

Visualization 1D

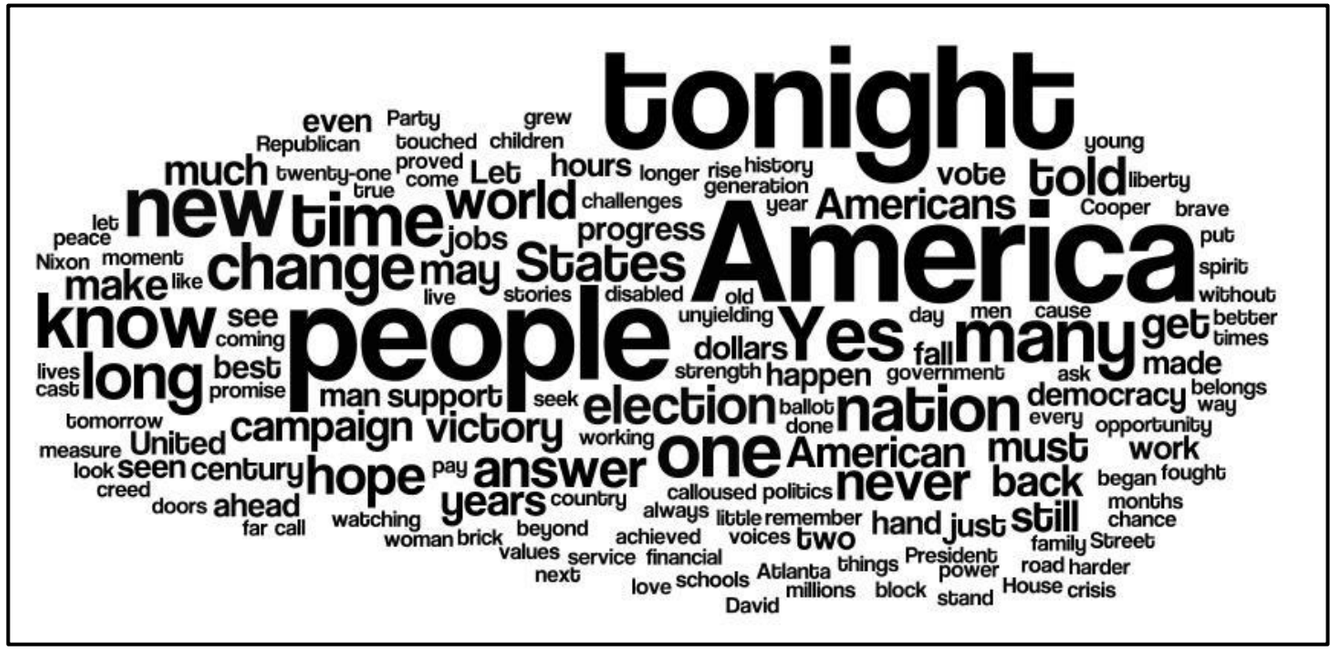
Word Cloud



Obama's 2010 speech

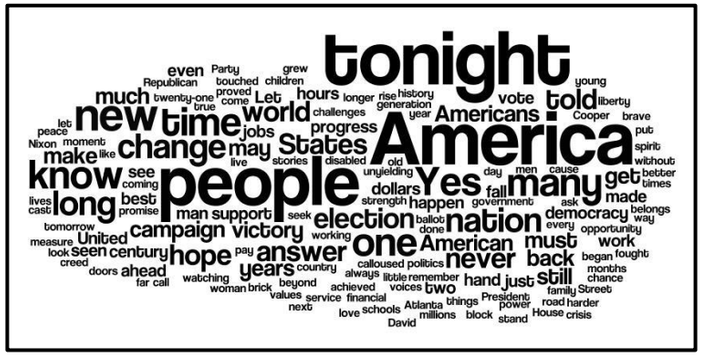
- Widely used in visualization for texts;
- Maps the number of occurrences in the word size;

Visualization 1D



What are the problems with a Word Cloud?

Visualization 1D



- Words with more letters stand out more than shorter words;
- Words in sequence are separated, which can completely change their meaning; Ex: “... I don’t want...”

Data Type

- Two-dimensional (2D): planar or cartographic data (maps).

Cities	Latitude	Longitude
Lund	-23.5489	-46.6388
Växjö	-22.9035	-43.2096
Malmö	-32.0332	-52.0986

[The eyes have it: A task by data type taxonomy for information visualization, Shneiderman 96]

Data Type

- Temporal: Data that evolves over time.

- Weather forecast:

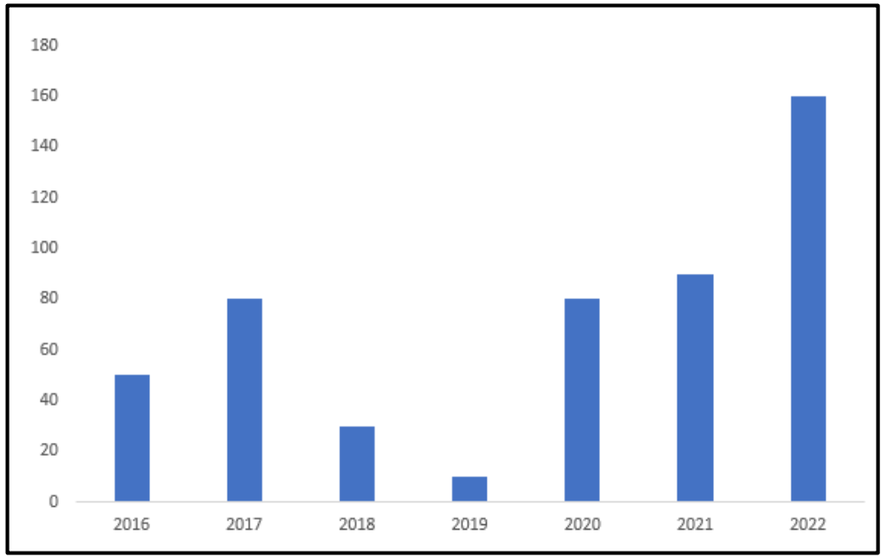
Temp. (ºC)	Date
32	18/11/2021
25	19/11/2021
13	20/11/2021
22	21/11/2021
19	22/11/2021
27	23/11/2021

[The eyes have it: A task by data type taxonomy for information visualization, Shneiderman 96]

Visualization 2D / T

- Bar chart

Year	Sells
2016	50
2017	80
2018	30
2019	10
2020	80
2021	90
2022	160

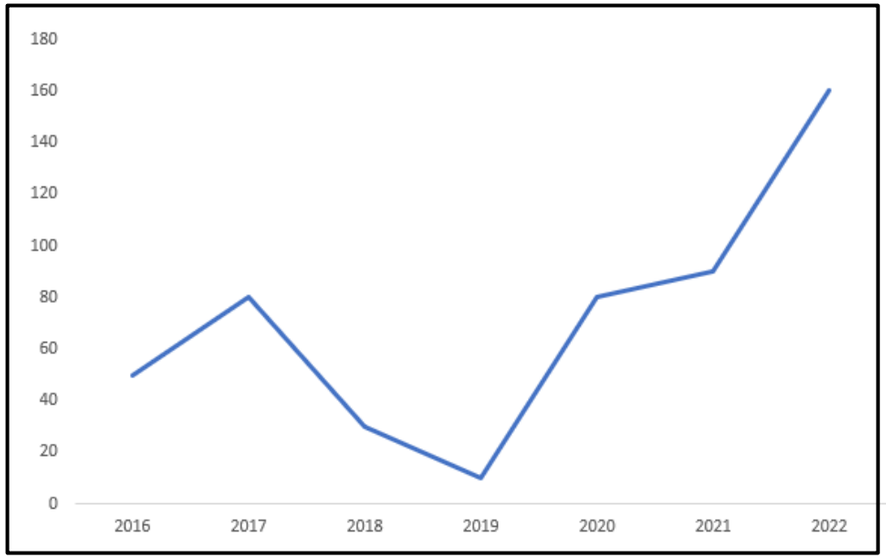


Works well for quantitative data.

Visualization 2D / T

- Line chart

Year	Sells
2016	50
2017	80
2018	30
2019	10
2020	80
2021	90
2022	160



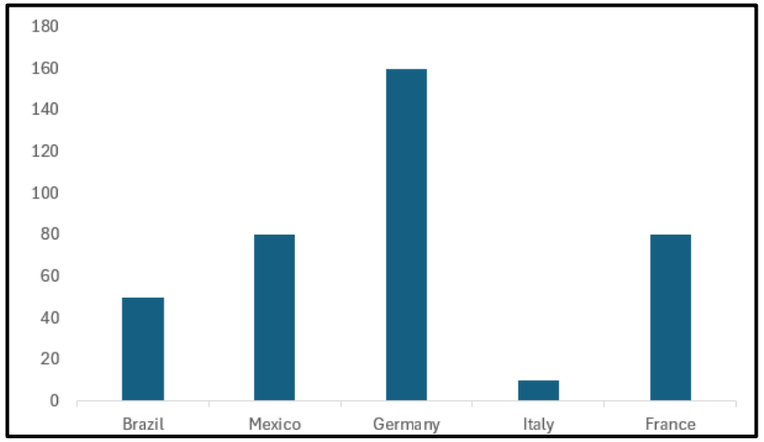
It is recommended for quantitative data, especially with time on the X axis.

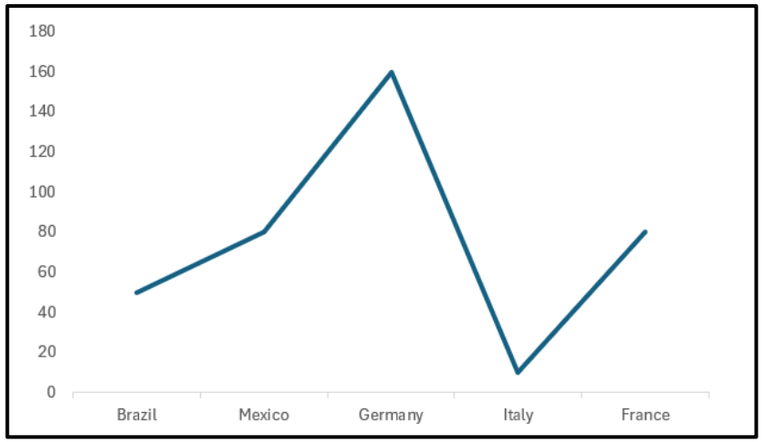
Better to identify small changes.

Visualization 2D / T

- Bar/Line chart

Country	Sells
Brazil	50
Mexico	80
Germany	160
Italy	10
France	80

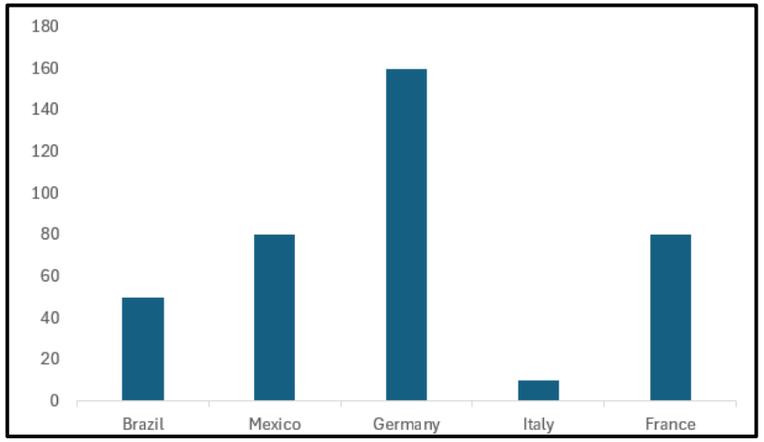


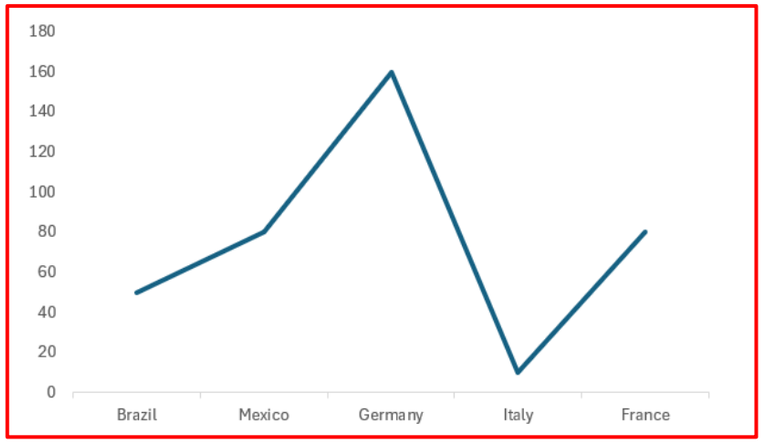


Visualization 2D / T

- Bar/Line chart

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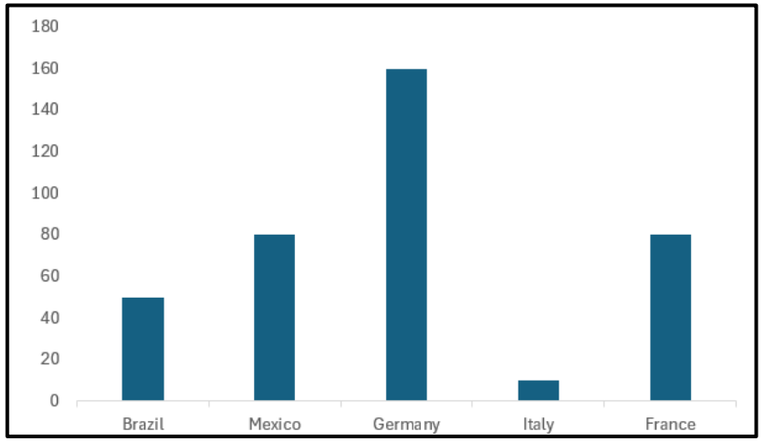


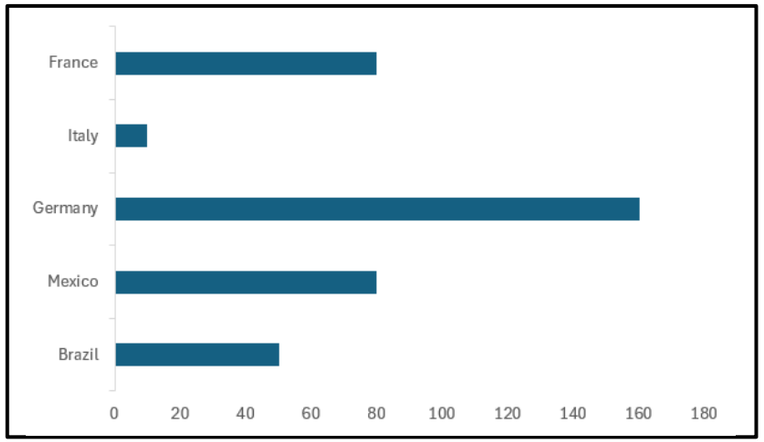
X

Visualization 2D / T

- Line chart

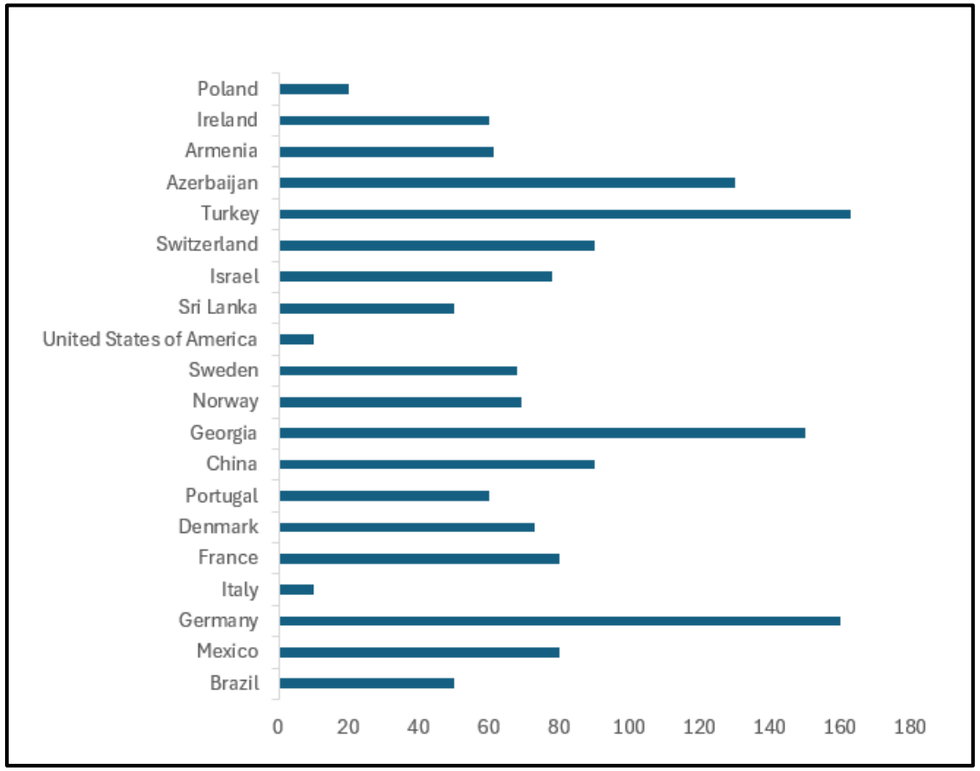
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Italy	10
France	80





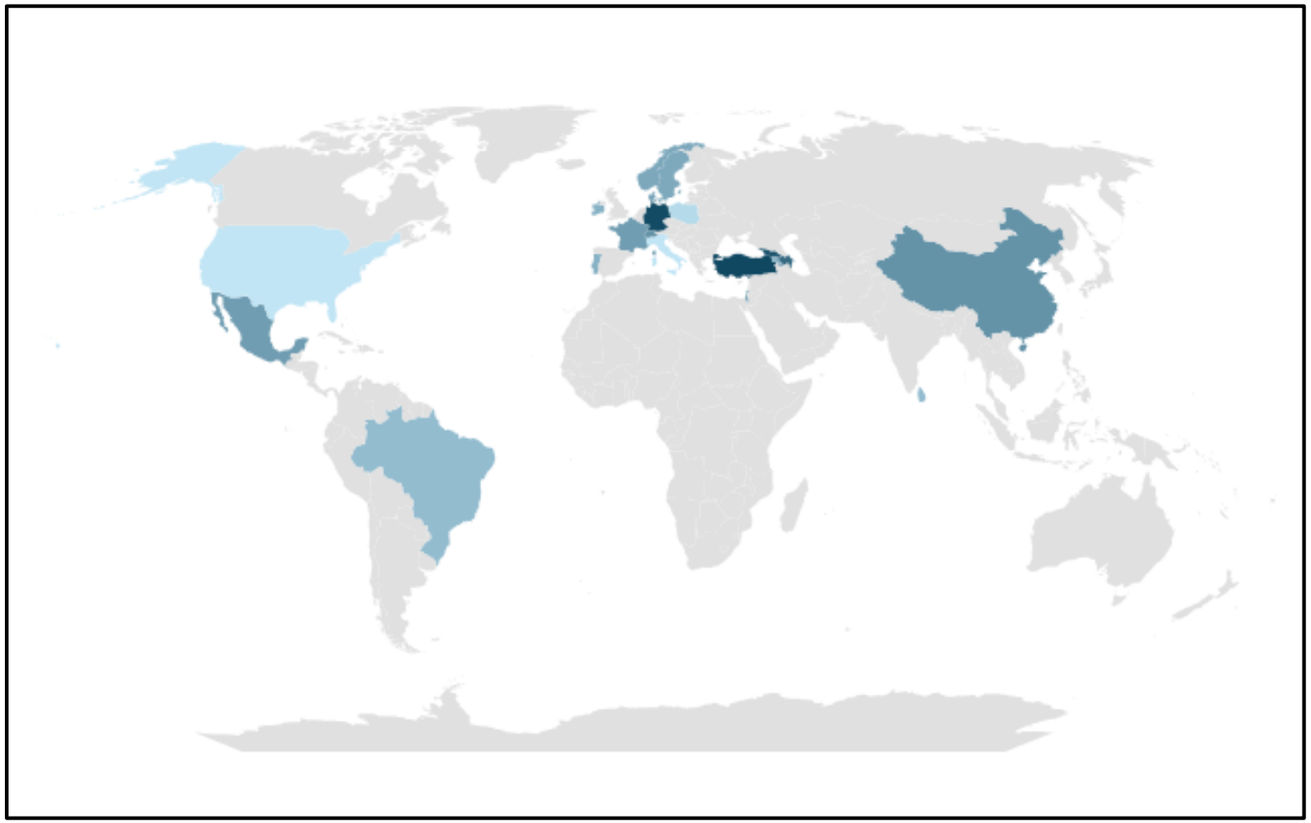
Visualization 2D / T

- Bar chart



Visualization 2D / T

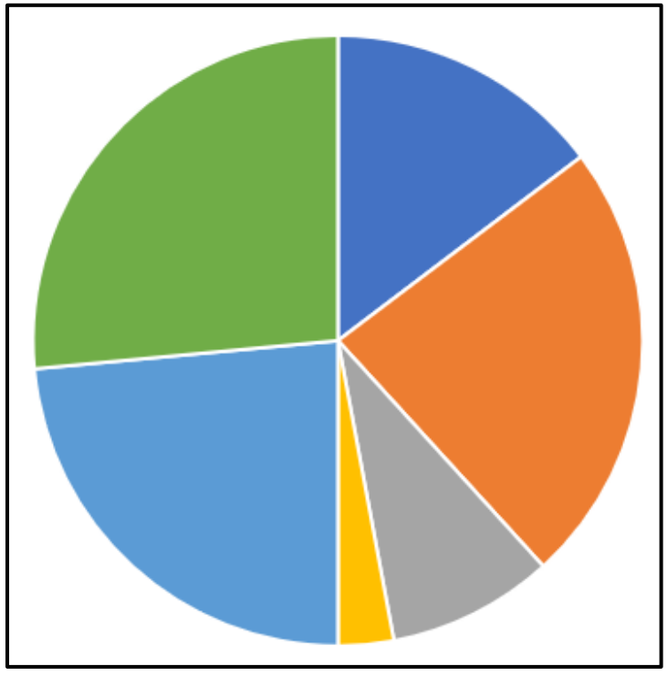
- Map



Alternative: maps mapping the number of sales based on color intensity!

Visualization 2D / T

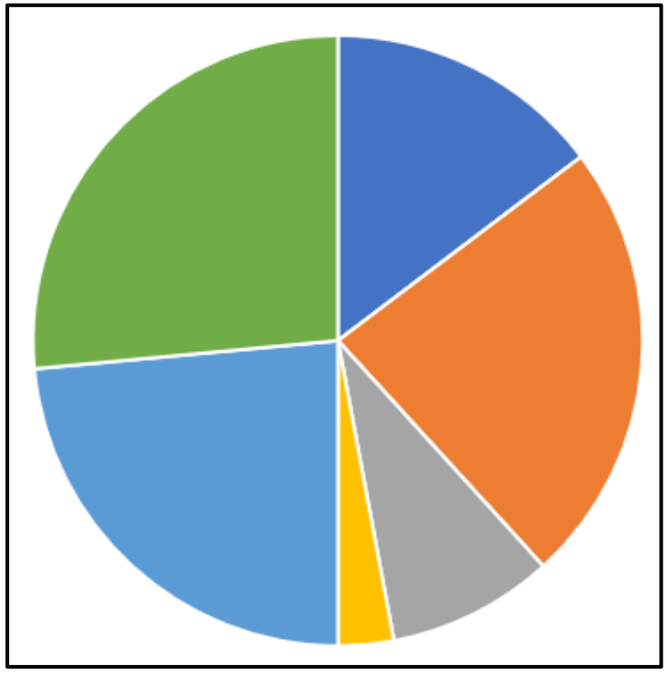
- Pie Chart



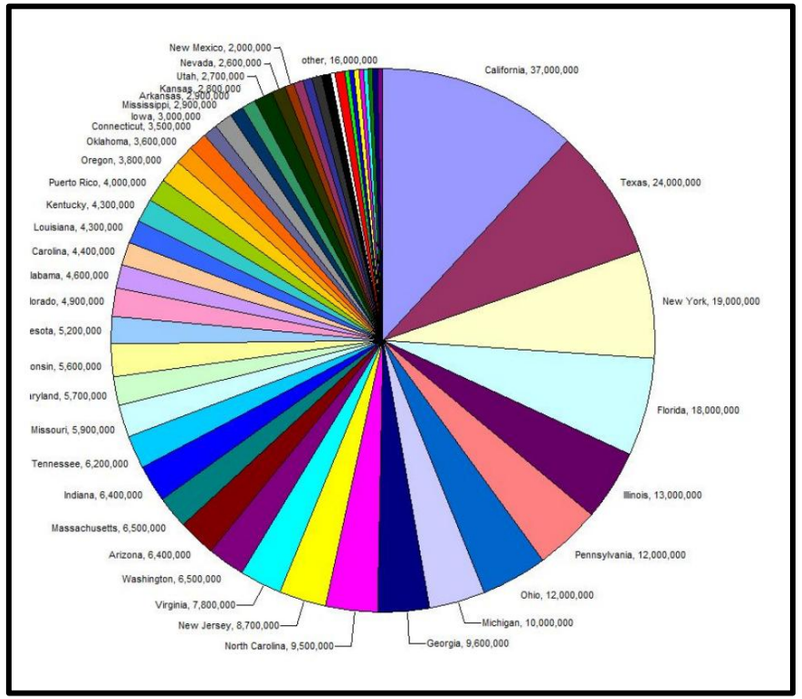
- It's hard to see small variations because of the different angles.

Visualization 2D / T

- Pie Chart

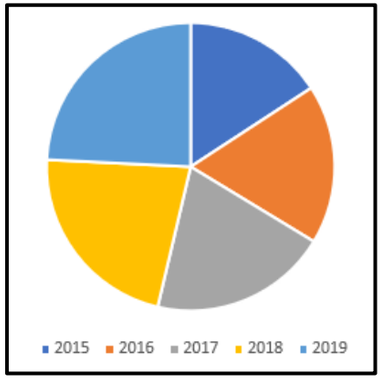


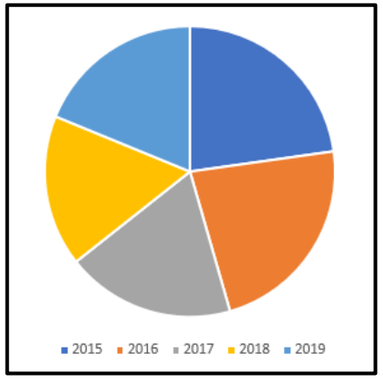
- It's hard to see small variations because of the different angles.
- So many different categories make it even harder to tell them apart!

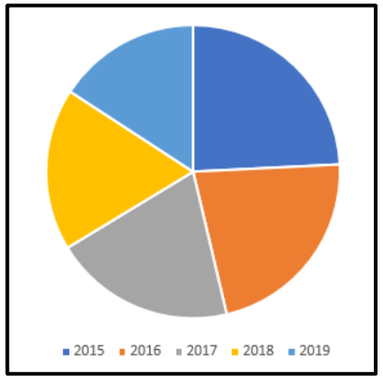


Visualization 2D / T

- Pie Chart



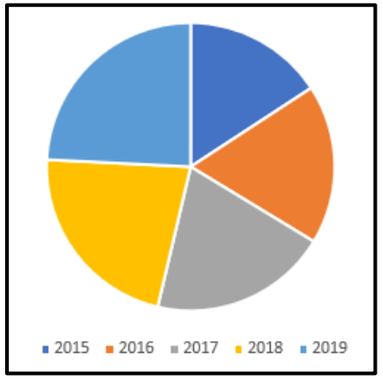


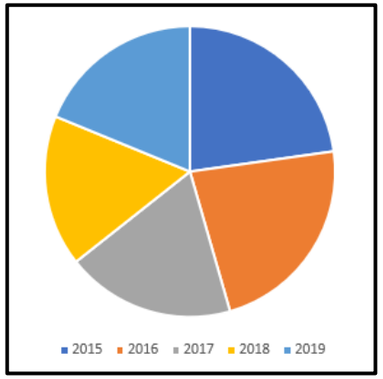


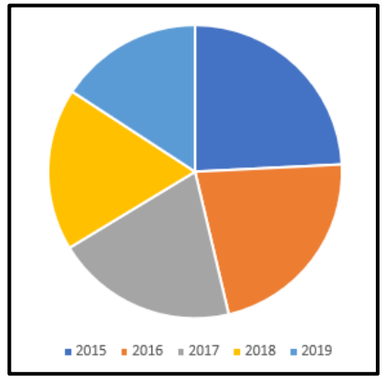
Which pie chart the data is growing, reducing, or stable over time?

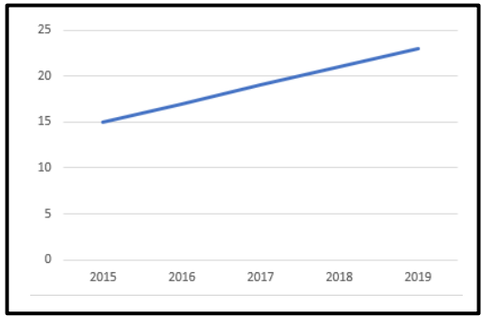
Visualization 2D / T

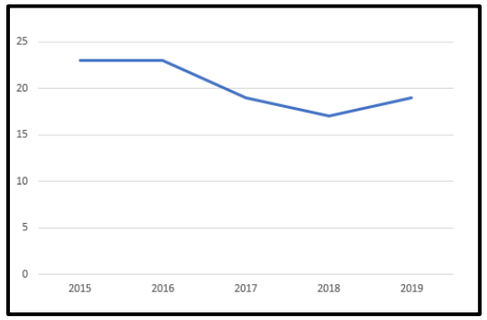
- Pie Chart

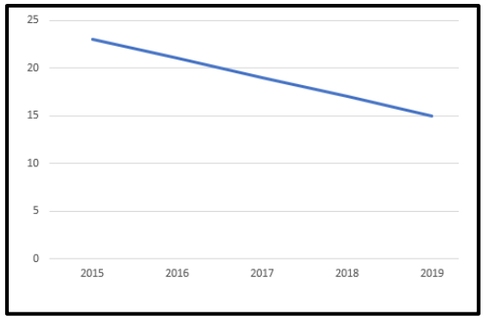












Data Type

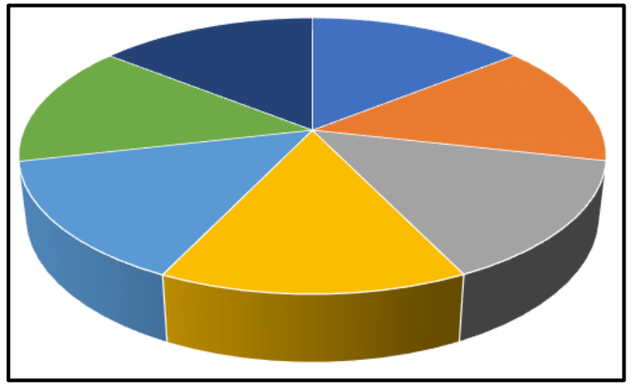
- Three-dimensional (3D): data related to volume and shape.

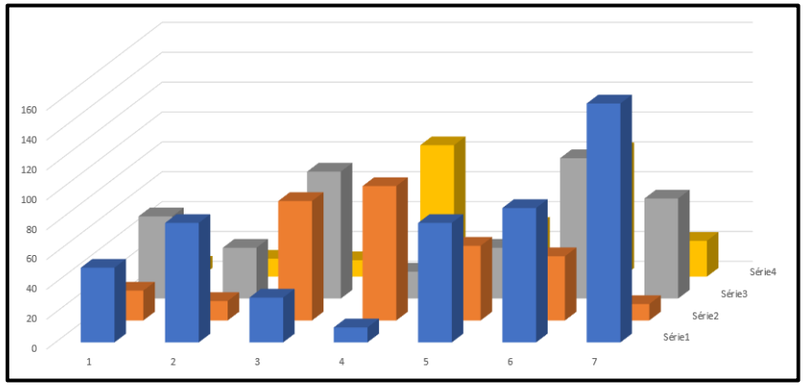


Object	Length	Width	Height
Box	16	3	11

[The eyes have it: A task by data type taxonomy for information visualization, Shneiderman 96]

Visualization 3D

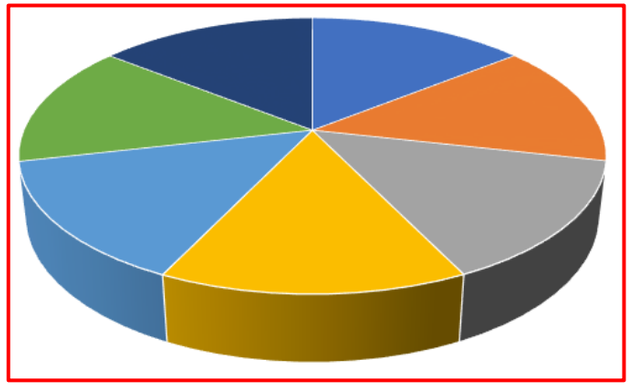




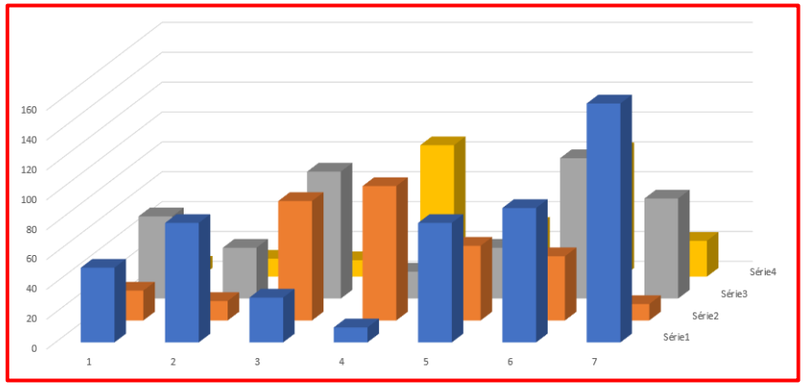
Any problems with 3D representations?

Visualization 3D

X

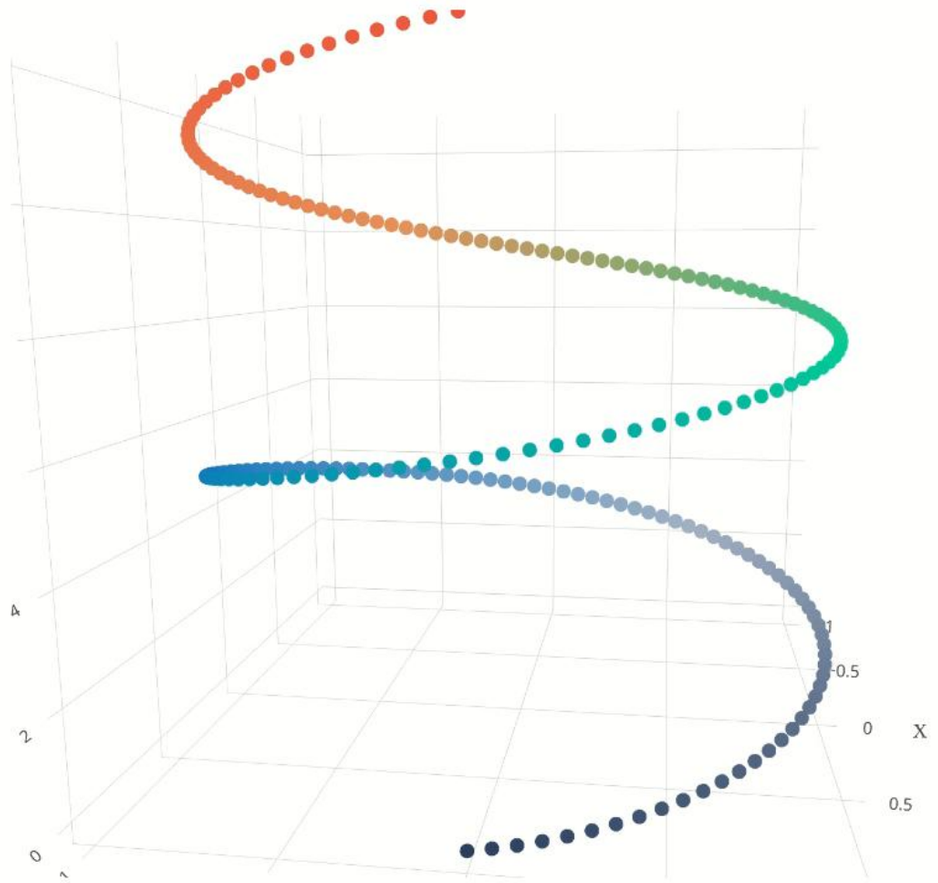


X



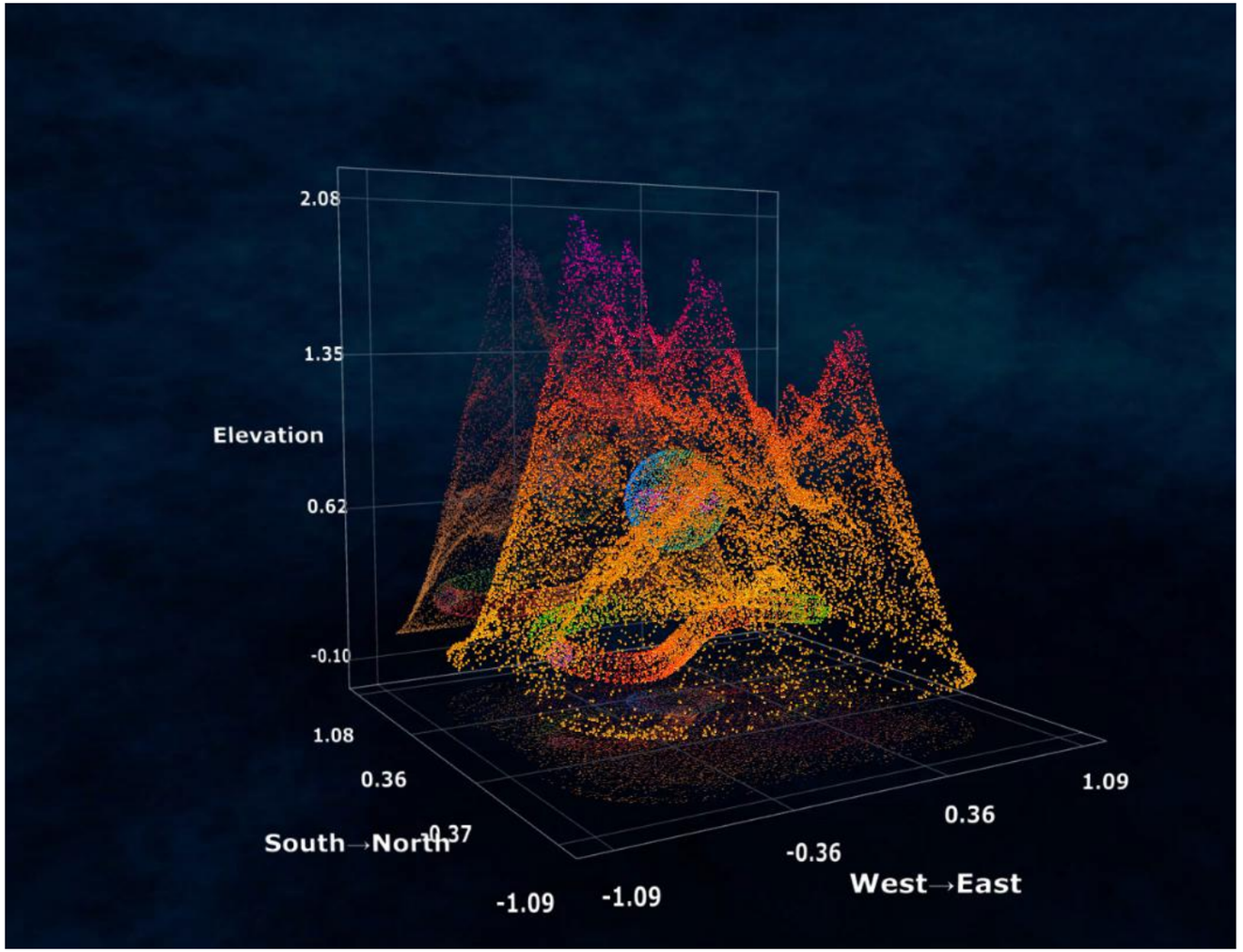
3D visualizations are not recommended without **interaction** or appropriate **devices**, as the perspective can distort the data.

Visualization 3D



https://plotly.com/chart-studio-help/make-a-3d-scatter-plot/

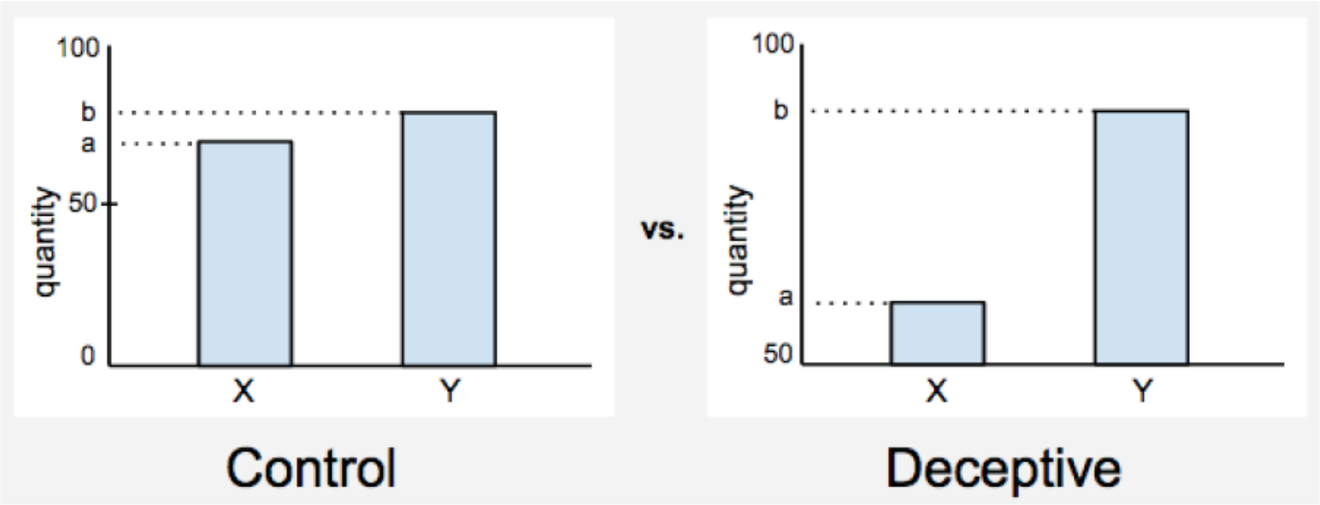
Visualization 3D



https://miabellaai.net/

Misleading Visualization

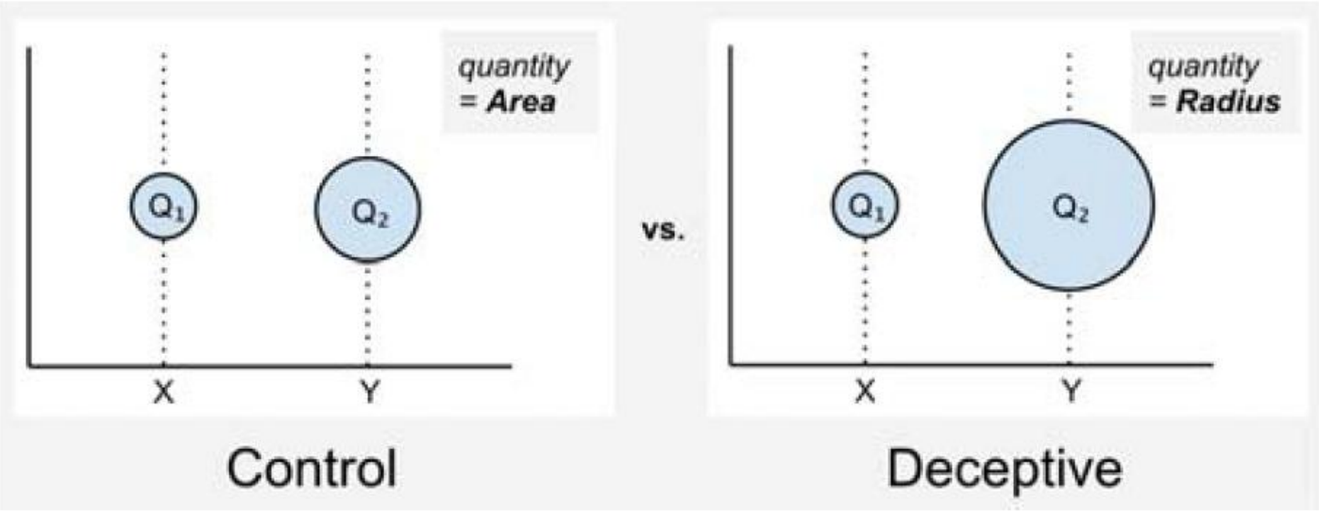
- Truncated Axis distortion
  - Leads to message exaggeration/understatement type of deception



How deceptive are deceptive visualizations? An empirical analysis of common distortion techniques

Misleading Visualization

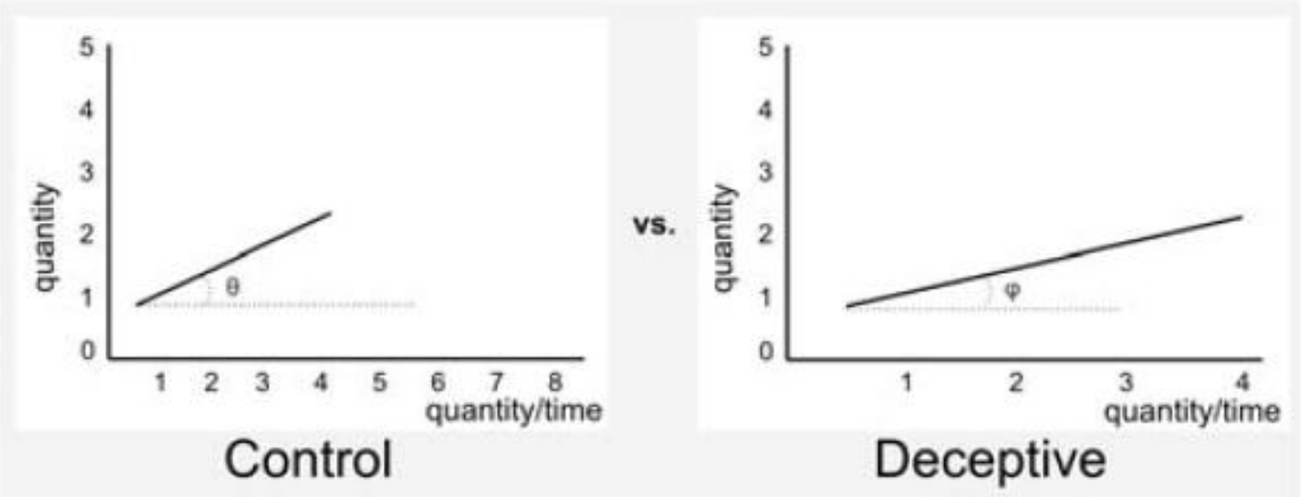
- Area as Quantity distortion
  - Leads to message exaggeration/understatement type of deception



How deceptive are deceptive visualizations? An empirical analysis of common distortion techniques

Misleading Visualization

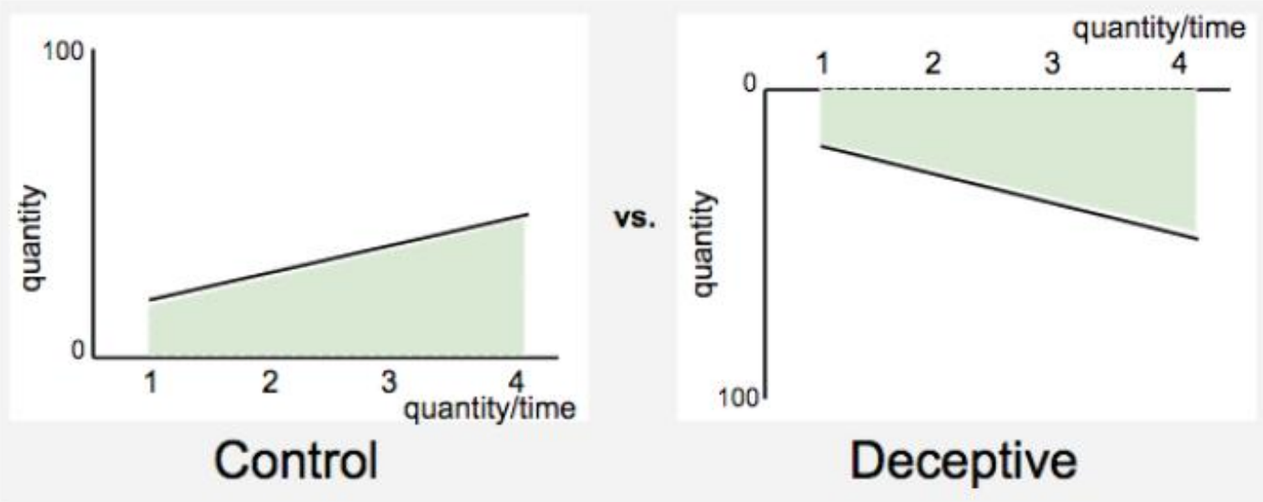
- Aspect Ratio distortion
  - Altered angle responsible for "rate of increase/decrease"
  - Leads to message exaggeration/understatement type of deception



How deceptive are deceptive visualizations? An empirical analysis of common distortion techniques

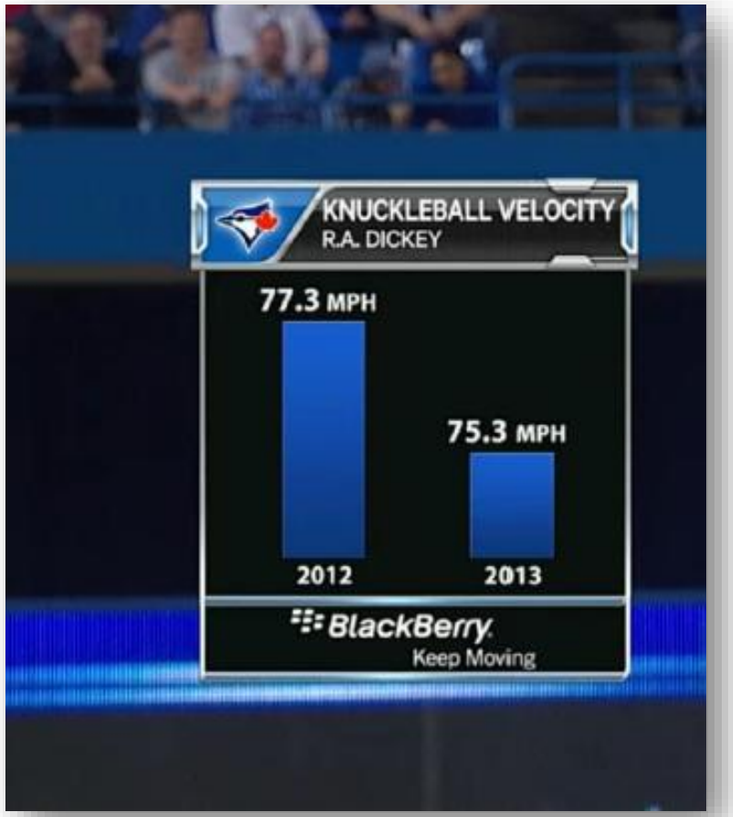
Misleading Visualization

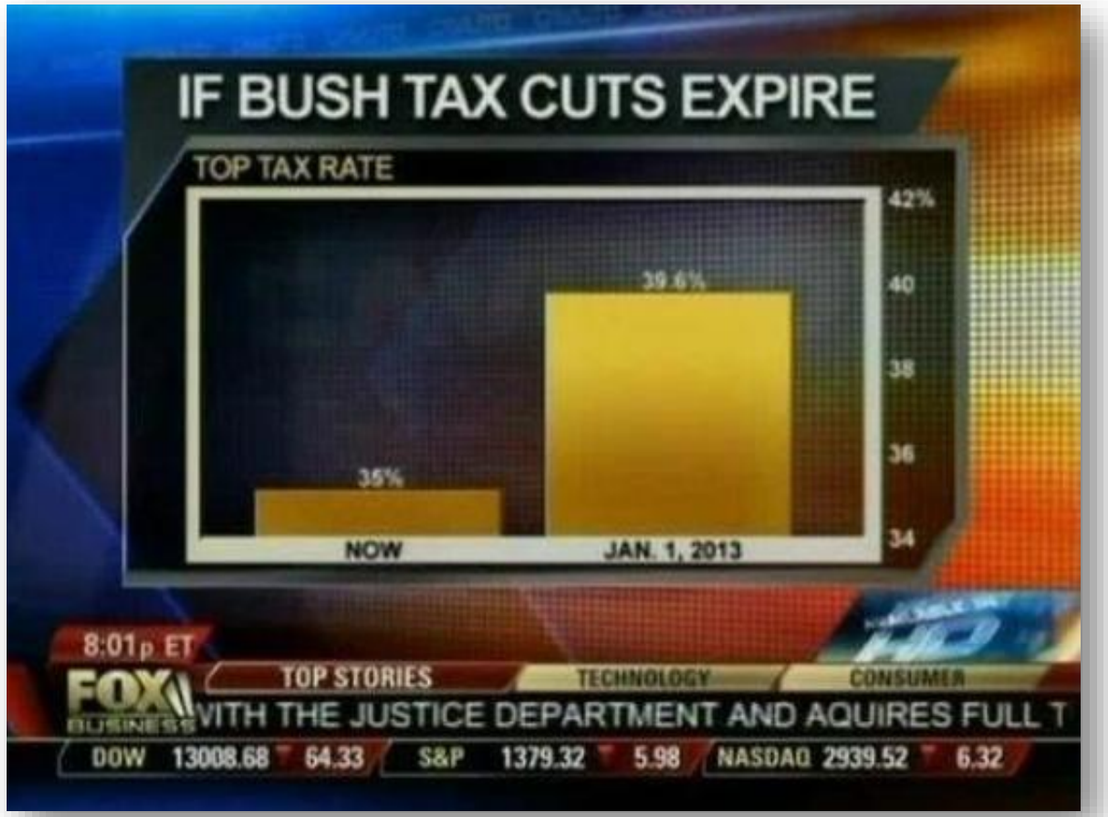
- Inverted Axis distortion
  - Leads to message reversal type of deception



How deceptive are deceptive visualizations? An empirical analysis of common distortion techniques

Misleading Visualization



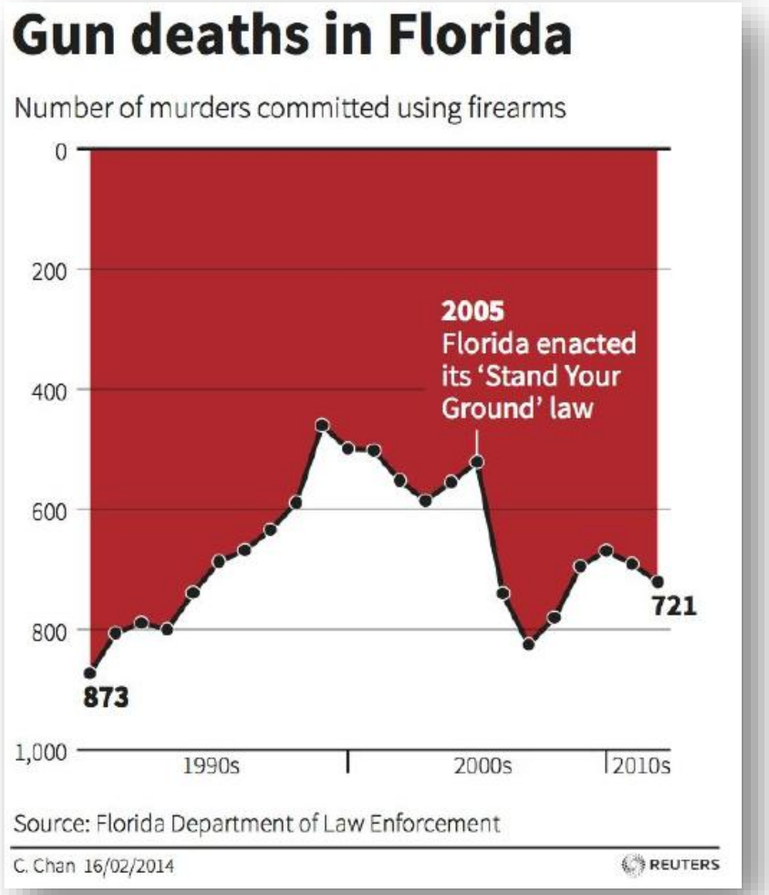


Misleading Visualization





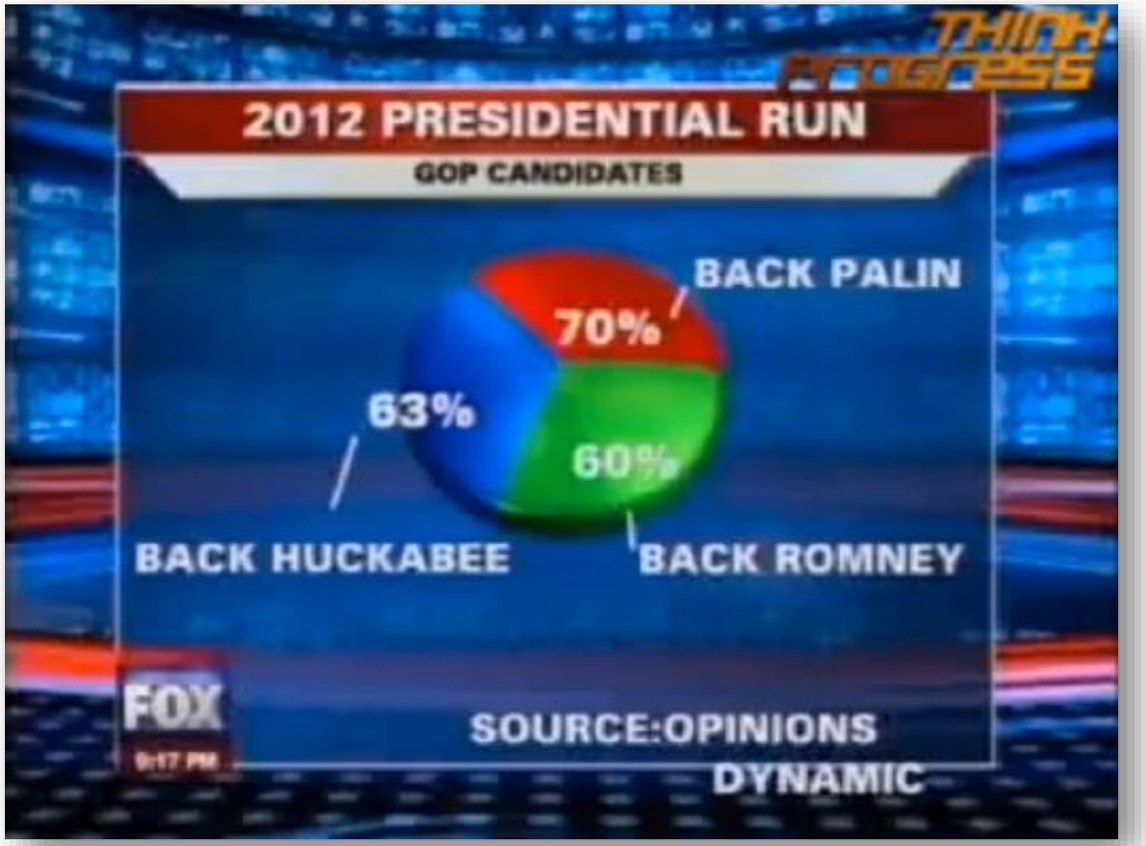
Misleading Visualization



Misleading Visualization

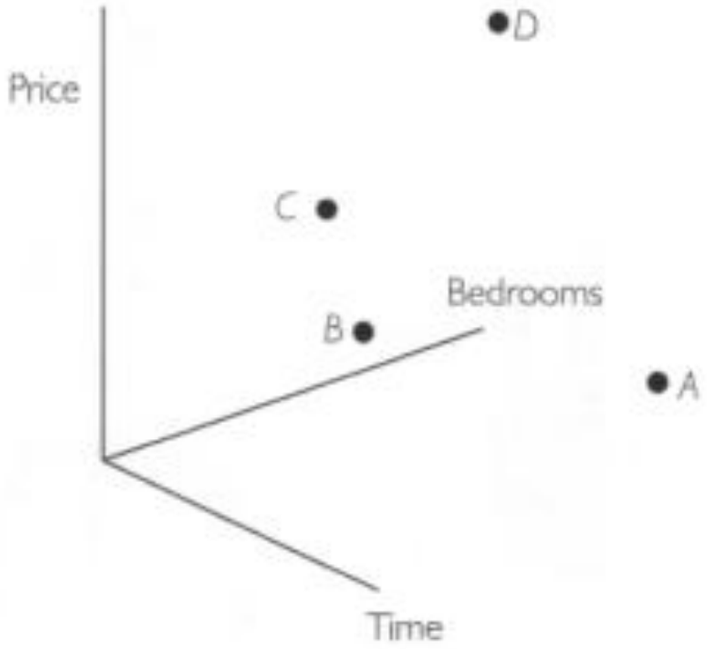


Misleading Visualization



Tables

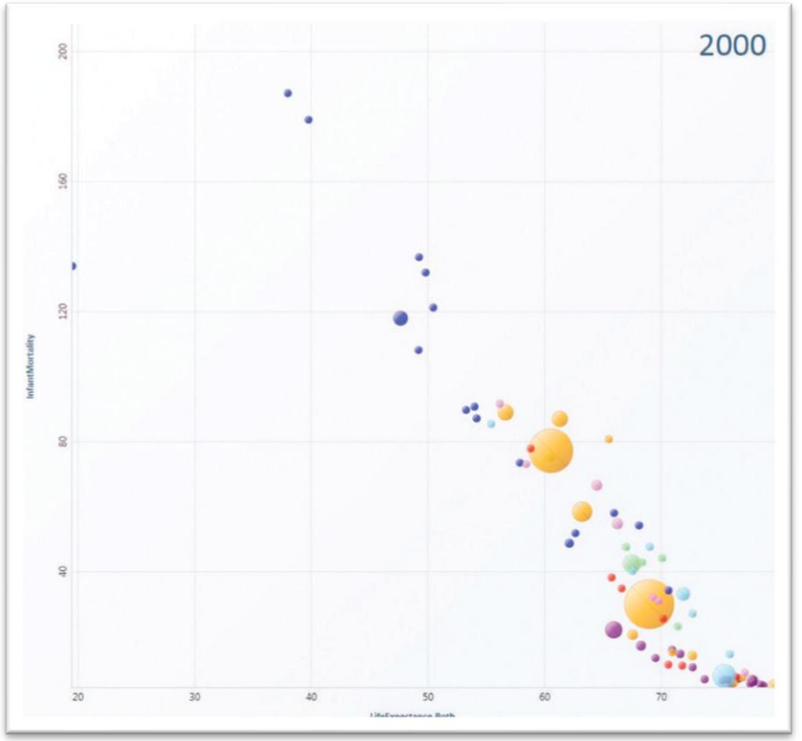
- **Visual Mapping of Tables:**
- Scatterplot?
  - Max. three dimensions
  - Even with 3D, what we are seeing is a 2D projection!



[Spe]

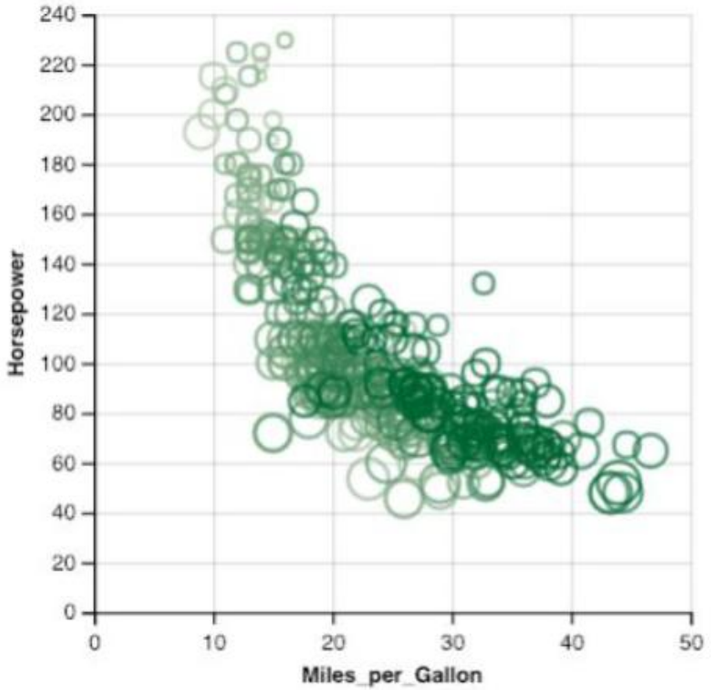
Tables

- **Visual Mapping of Tables:**
- Scatterplot?
  - Max. three dimensions
  - Even with 3D, what we are seeing is a 2D projection!
- Possibility: use different marks and channels
  - Color
  - Area

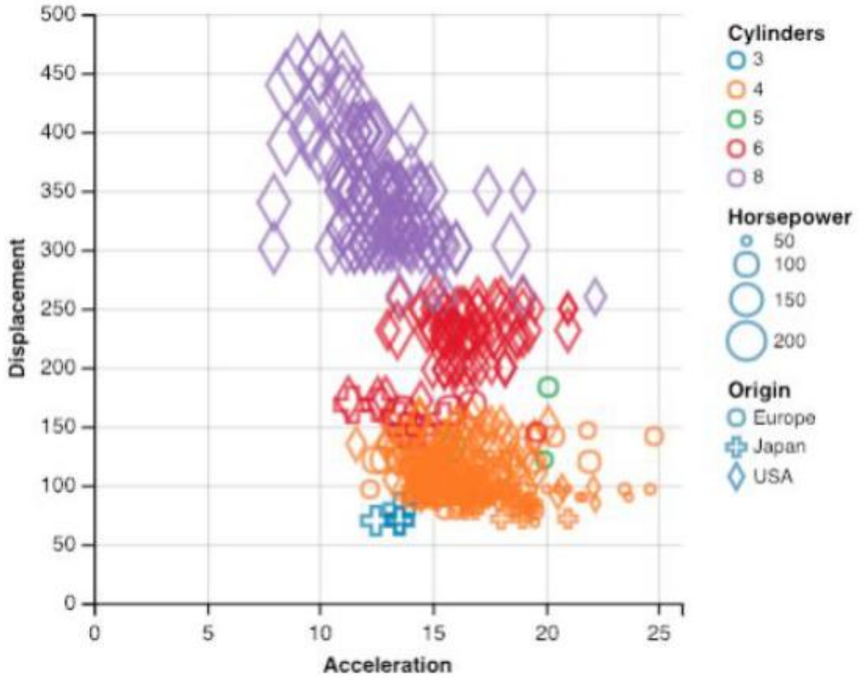


[Mun]

Tables



2-dimensional
+ 1 dimension (size)



2-dimensional
+ 2 dimension
(shape + color)

[Inspired by Benjamin Bach's lecture]

Tables

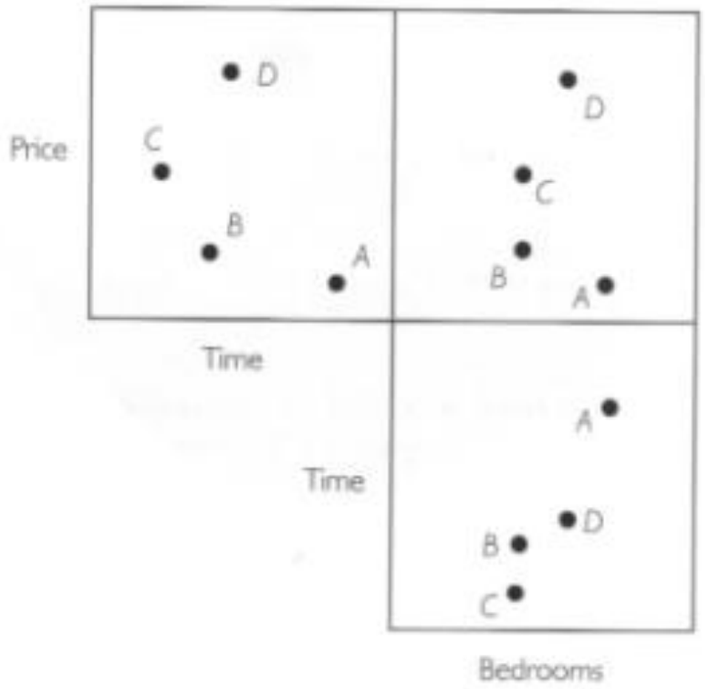
- **Visual Mapping of Tables**
  - Another solution would be projections along all three axes



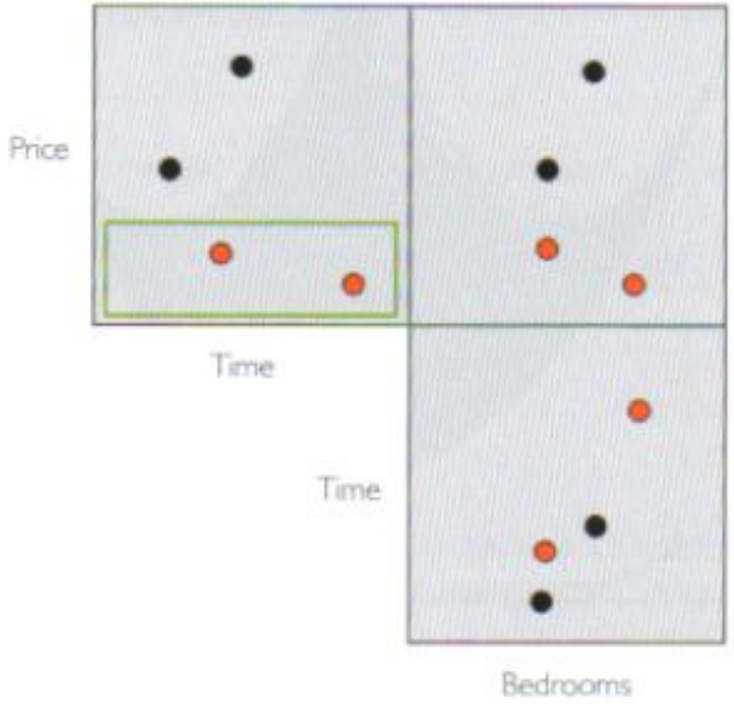
[Spe]

Tables

- Scatterplot Matrix



Scatter Plot Matrix



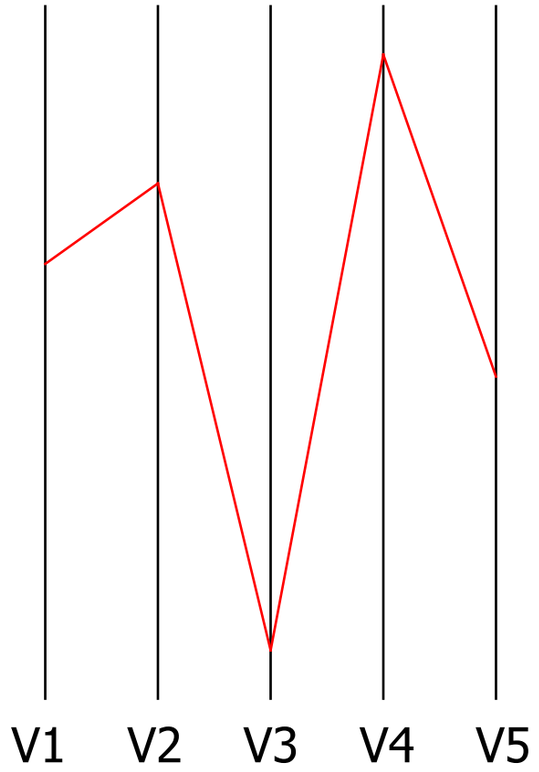
Scatter Plot Matrix with brushing

[Spe]

Axis-based

Parallel Coordinates

- If we have *n* variables, then we can draw *n* parallel axes and enter the values on the corresponding axes
- Each axis is scaled on the minimal/maximal range of all occurring values
- Each data object is represented by a polygon through all axes
- Well-known and multiple used technique

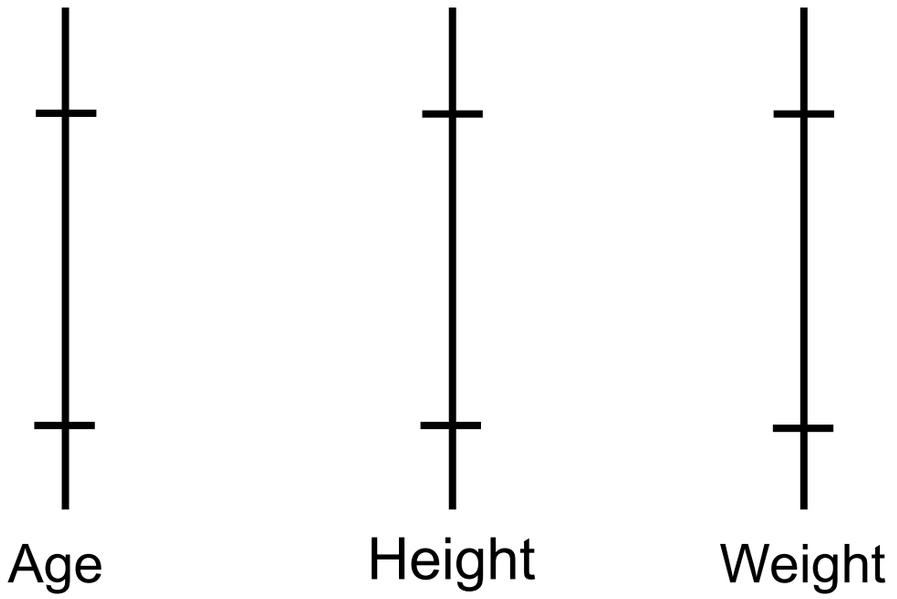


[A. Inselberg, "Multidimensional Detective", InfoVis '97, 1997]

[Inspired by J. Stasko's course]

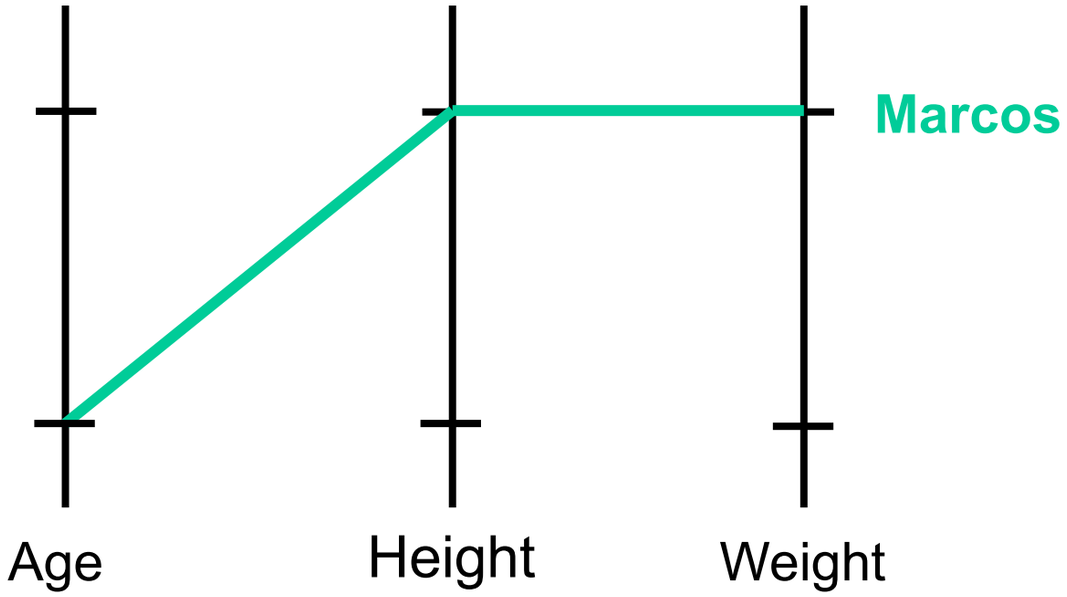
Parallel Coordinates

Users	Age	Height	Weight
Marcos	16	1.69	72
Maria	41	1.55	64



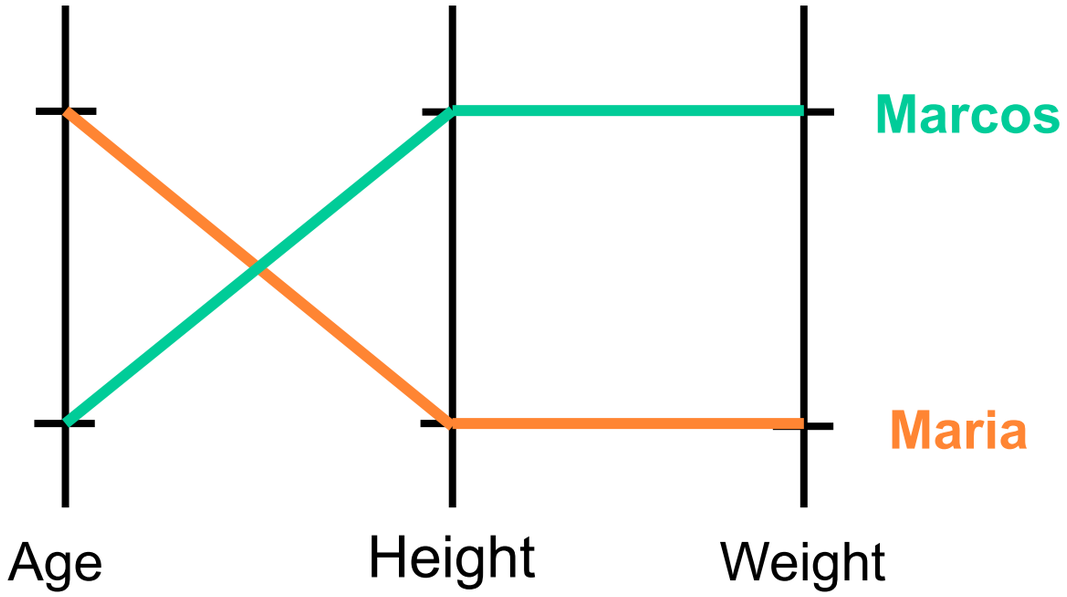
Parallel Coordinates

Users	Age	Height	Weight
Marcos	16	1.69	72
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Parallel Coordinates

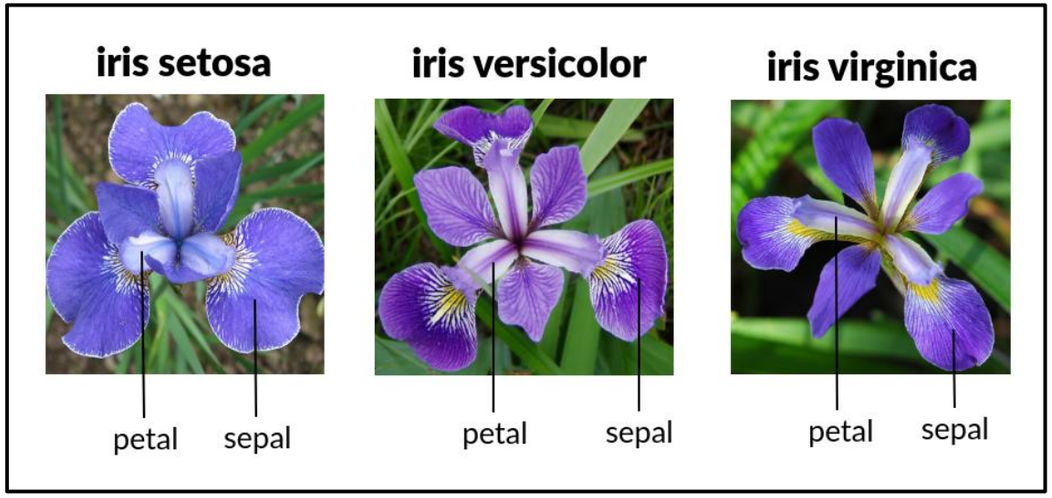
Users	Age	Height	Weight
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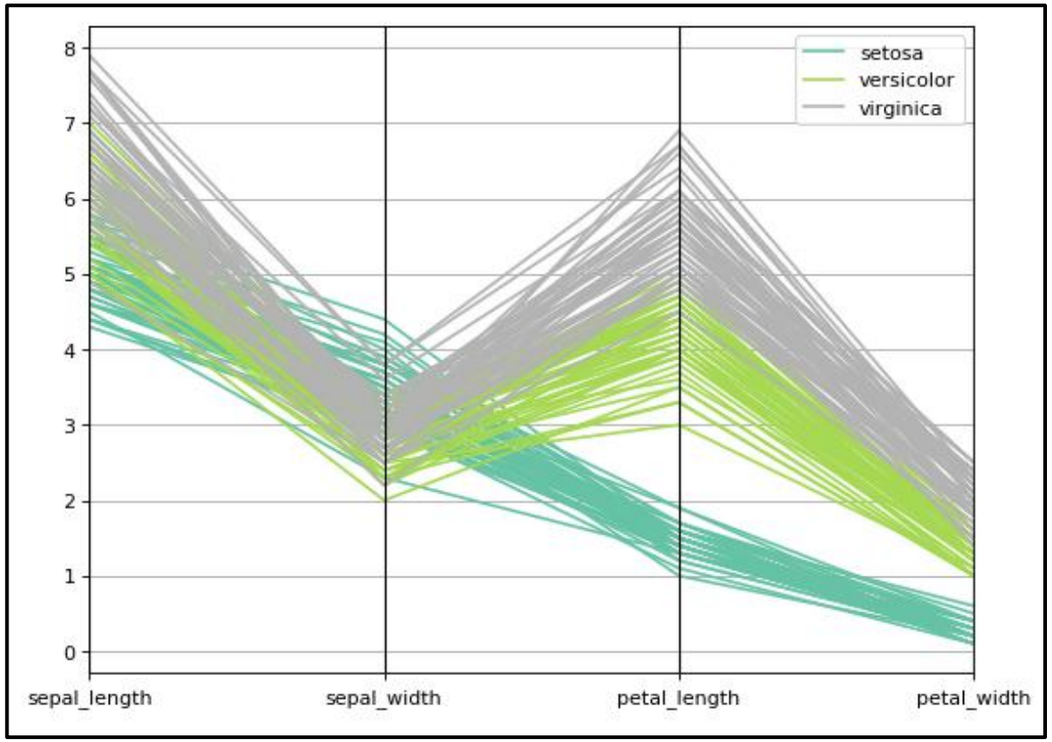


Parallel Coordinates

Iris *dataset*:

- Length and width of petals and sepals of three types of iris;

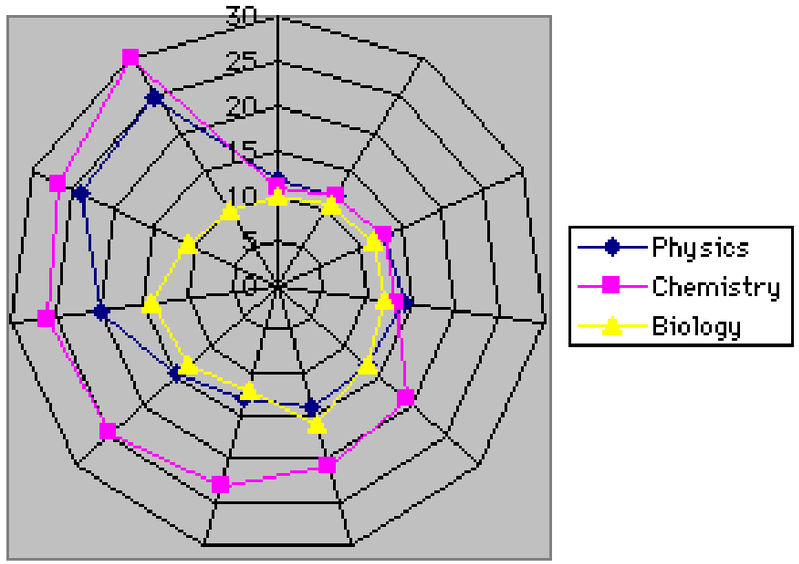


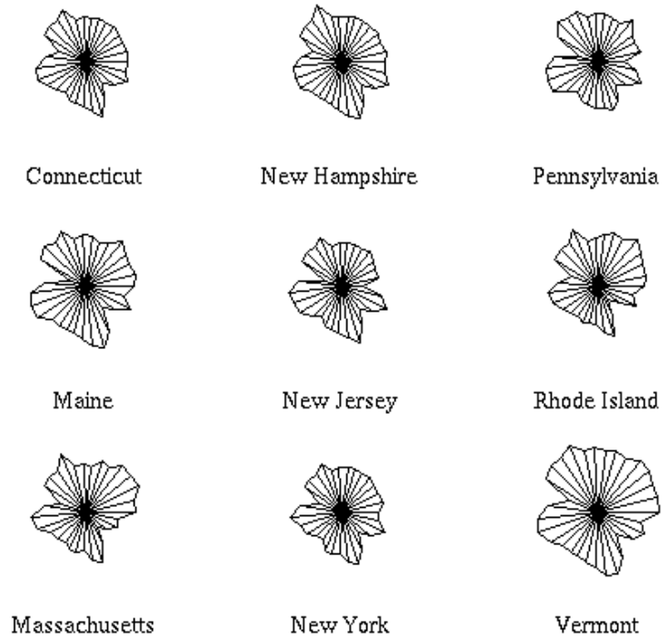


Axis-based

Star Plots

- Similar approach as Parallel Coordinates



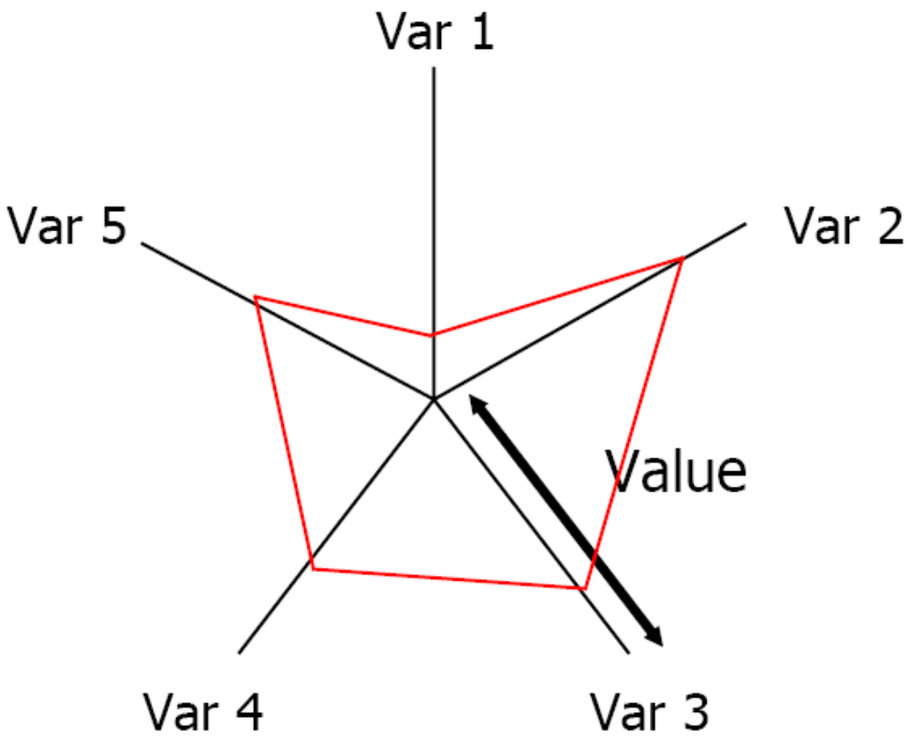


[Inspired by J. Stasko's course]

Axis-based

Star Plots

- Firstly, draw the axes like a star
- Each “spoke” codes a variable value
- One can build a single star for each data record, but also one star which contains/represents all records (→ overlapping problem)

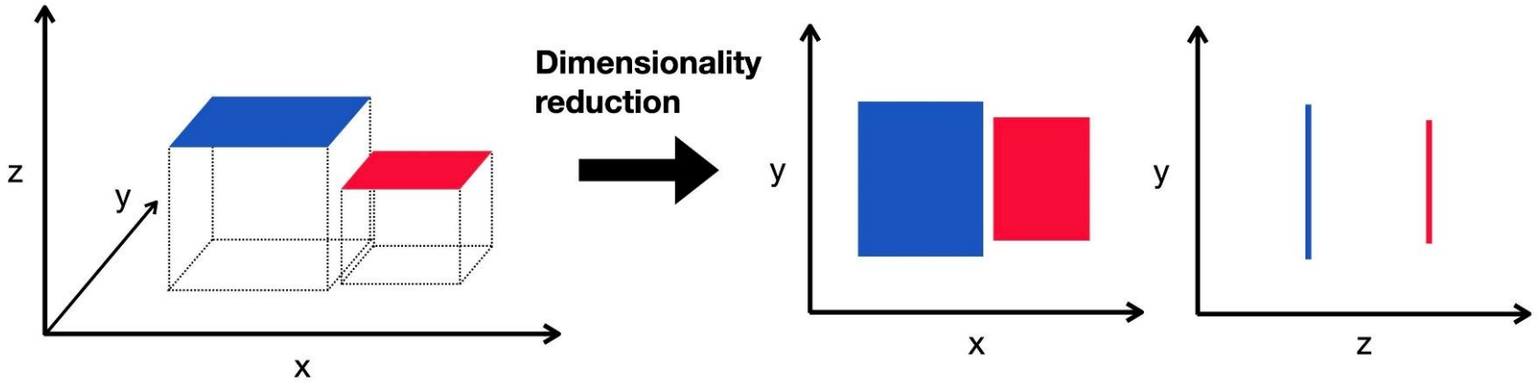


[Inspired by J. Stasko‘s course]

Dimensionality Reduction

- A technique to simplify complex datasets by reducing the number of variables while preserving important information.

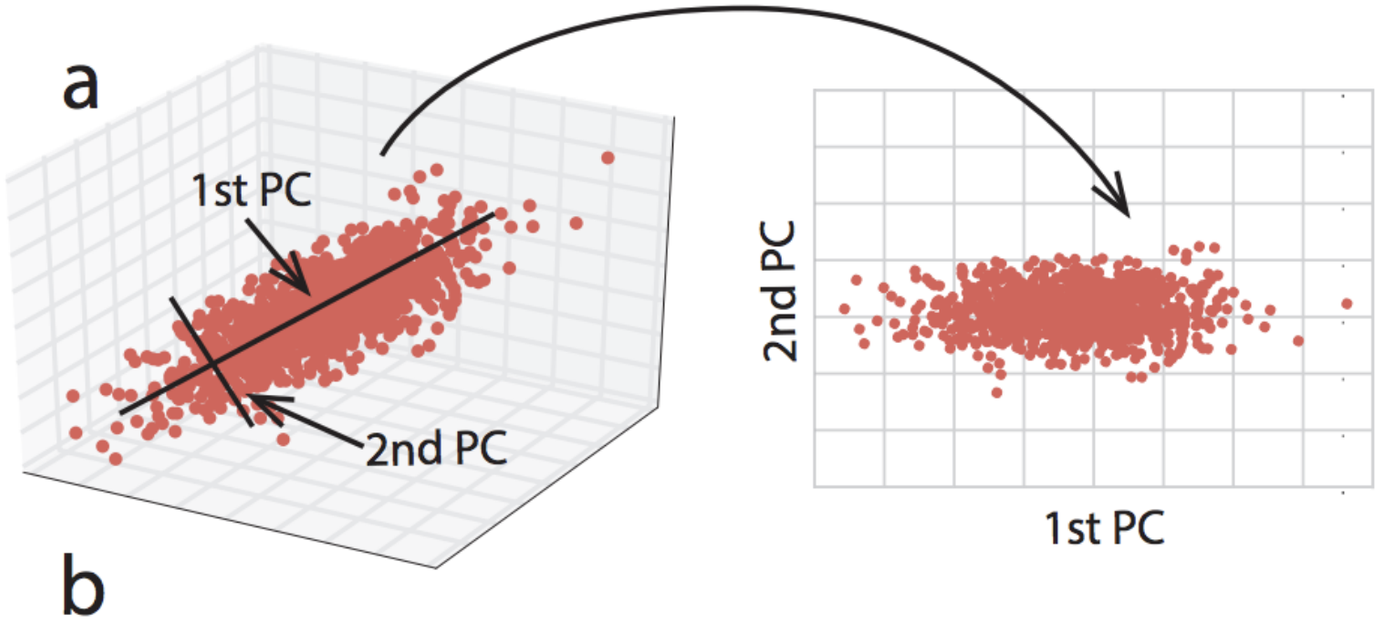
- Dimensionality reduction helps make data easier to visualize and analyze, especially when working with high-dimensional datasets.



https://www.sc-best-practices.org/preprocessing_visualization/dimensionality_reduction.html#id140

Dimensionality Reduction

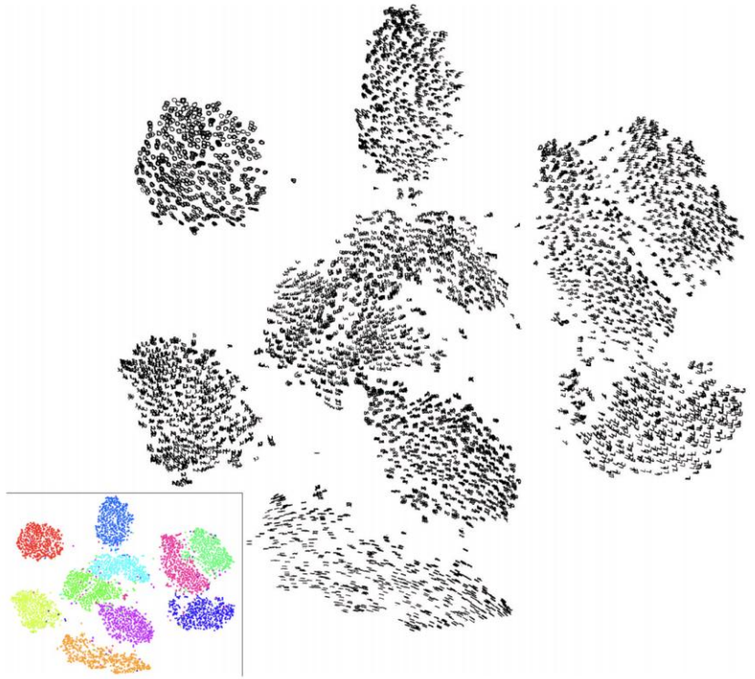
- Example: Principal Component Analysis (PCA)



Dimensionality Reduction

- T-SNE (van der Maaten & Hinton, 2008)





http://projector.tensorflow.org

Further Reading

- **[Mun]** T. Munzner. Visualization Analysis and Design. 1st Edition, A K Peters/CRC Press, ISBN 978-1466508910, 2014.
  - Chapter 2: What: Data Abstraction

